MAGNETOM

Installation Instructions System

Magnet / In-Transit Filling

Document Version

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1.1 Prerequisites for Filling

1.1.1 General information

Only Siemens trained and accredited engineers and approved third parties are allowed to perform this work.

1.1.1.1 General

During transportation, magnets are sometimes held in a **storage facility.**

Magnet **helium levels** are dependant on the duration of the **transport time.** After leaving the factory. a magnet has typically **30 days before the helium will reach zero.** During this storage time, the magnets can be **re-filled before onward transportation** to the customer site.

On arrival at the storage facility, the **magnet's general condition** must be checked. This should be performed by the transport / shipping company.

If the **helium level is low,** sufficient helium must be ordered and a scheduled fill arranged with a **third party company.**

Fig. 1: Kilian pallet

(1) Packing & Unpacking Instructions for the Magnet

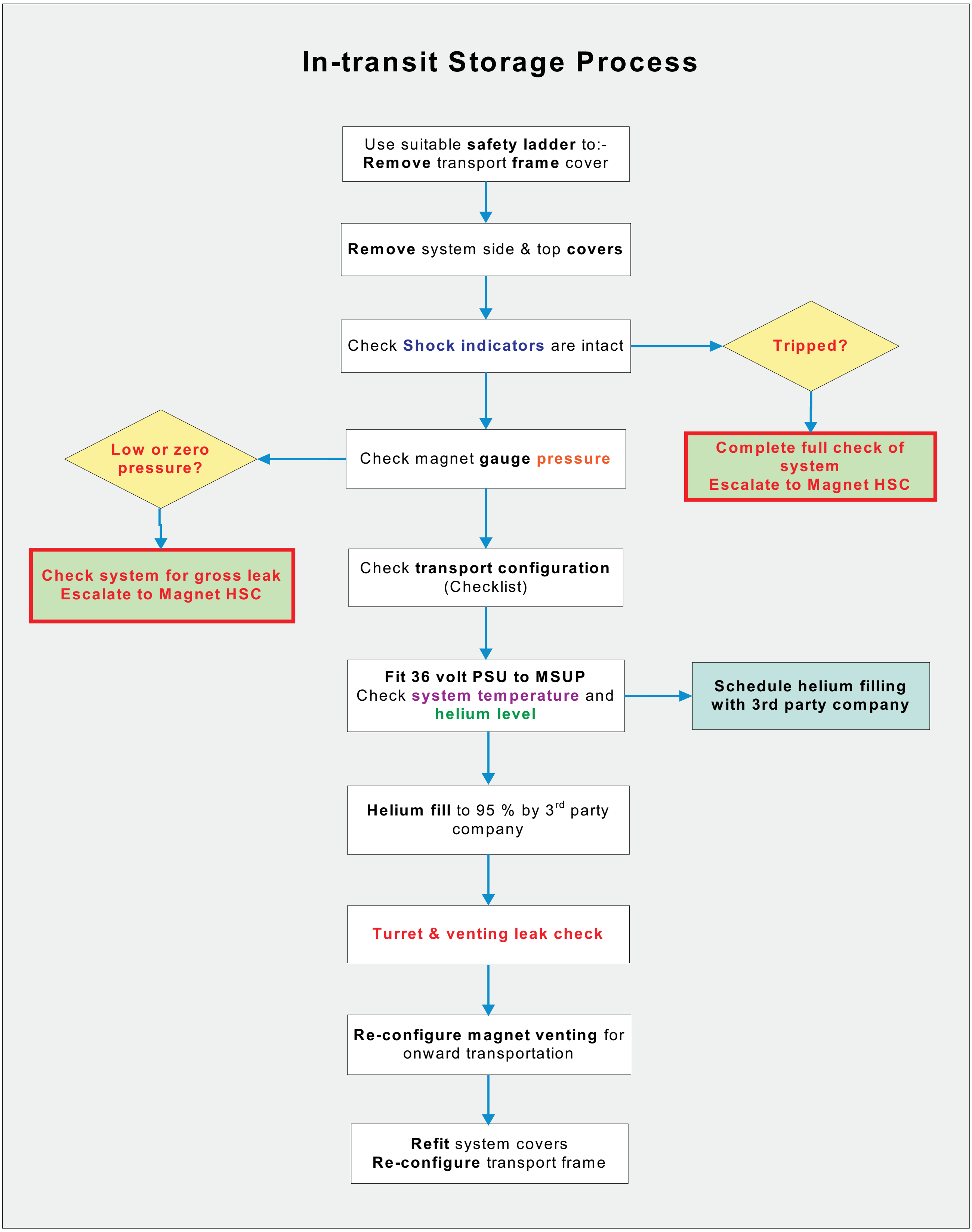
Tab. 1 Magnet Weights

OR98	OR99	OR103	Comments
3040 kg	5770 kg	5350 kg	Without helium
3200 kg	5820 kg	5500 kg	With 95% helium
3300 kg	6920 kg	6600 kg	System and transport pallet

Tab. 2 Magnet hold times

OR98	OR99	OR103	Cold head valve
30 days	30 days	30 days	Open
15 days	20 days	20 days	Closed

Fig. 2: OR98 / 99 / 103 In-transit Storage Process



Estimated completion time and number of persons

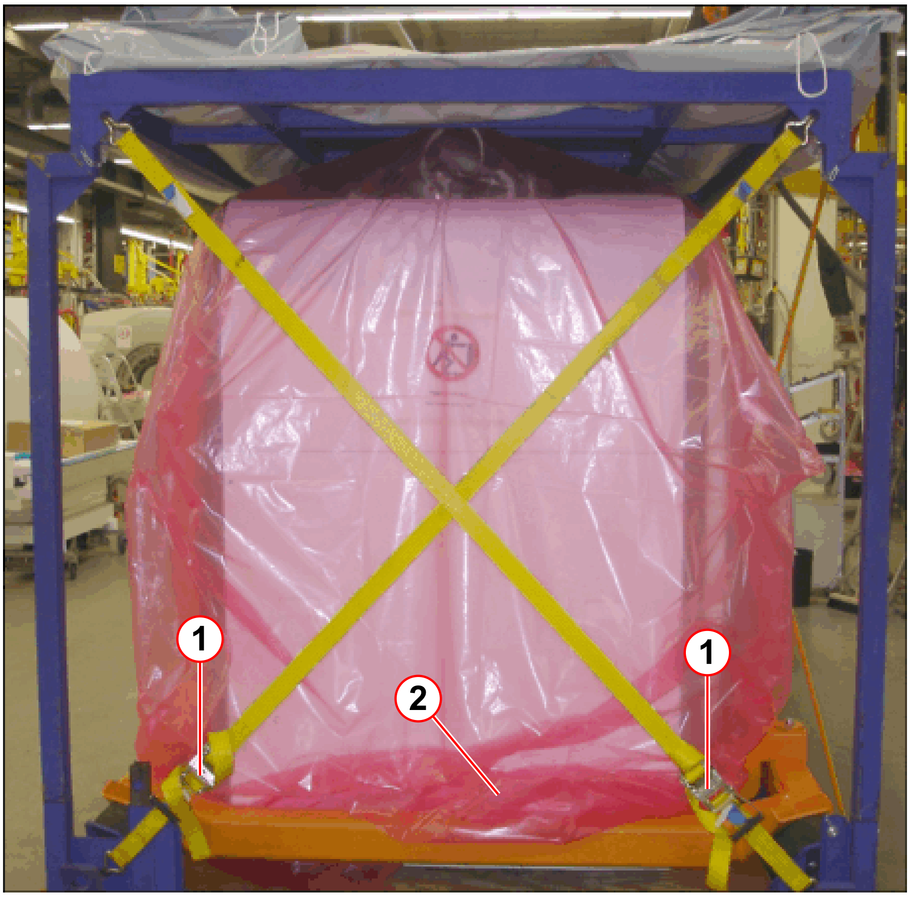
Tab. 3 Completion time

Worksteps	Time (Minutes)	Persons
Preparation	60 min	1
Helium filling	60 - 150 min	1
Final filling steps	60 min	1
Onward transport checks	60 min	1

Accessing the transport frame

Fig. 3: Kilian pallet with cover

Fig. 4: Access the (service end) magnet covers



(1) Remove straps
(2) Protective plastic cover

- Undo and remove the yellow **securing straps**.
- **Carefully** remove the **protective plastic covering**.

Pallet roof access to the magnet venting

Fig. 5: View of magnet venting (from pallet roof)

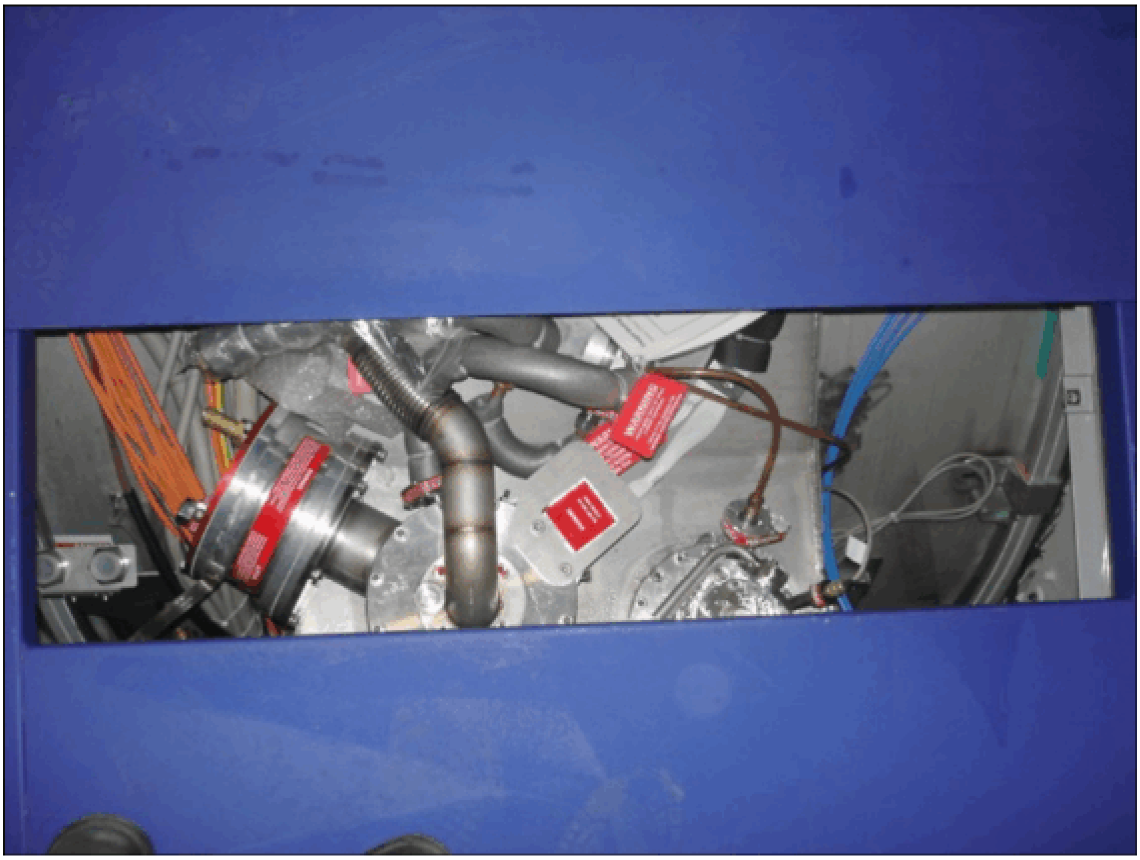
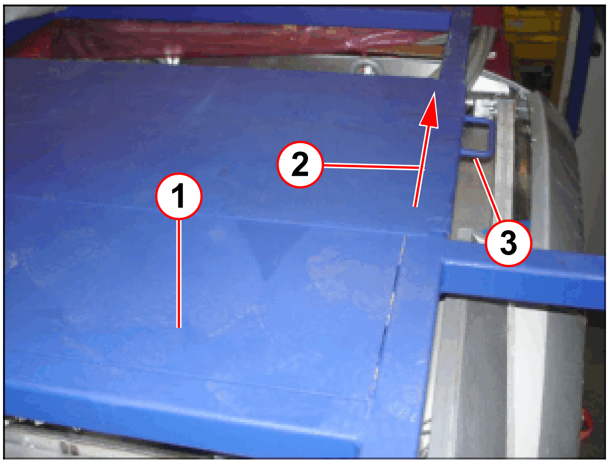
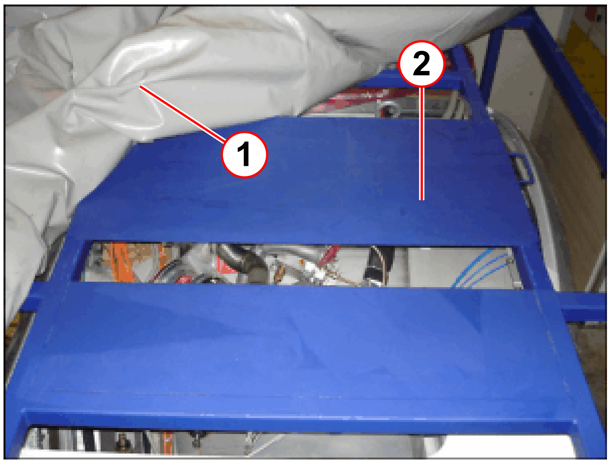


Fig. 6: Pallet roof cover (closed)



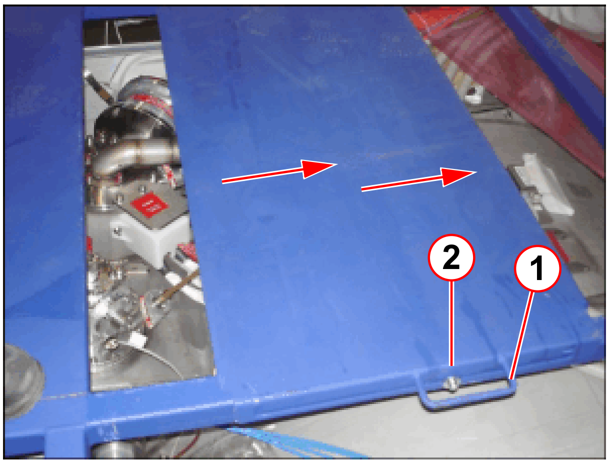
(1) Fixed panel
(2) Sliding panel
(3) Handle

Fig. 7: Remove the plastic roofing material



(1) Heavy duty plastic roofing material
(2) Sliding roof section

Fig. 8: Slide the roof panel to gain access



(1) Handle
(2) Lock the panel in place

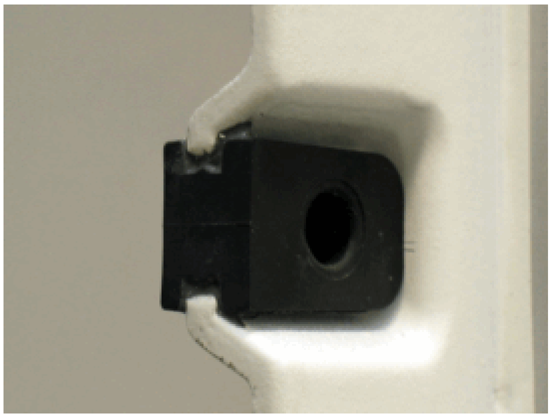
- Pull the handle to slide the roof panel away (pos 1).
- Lock the panel in place (pos 2).

Removing the magnet covers

- To **access the MSUP,** the **service side covers** (top and bottom) must be removed.
- Removing the magnet covers requires 1 person.
- The covers are mounted with screws and partly equipped with **snap locks.**

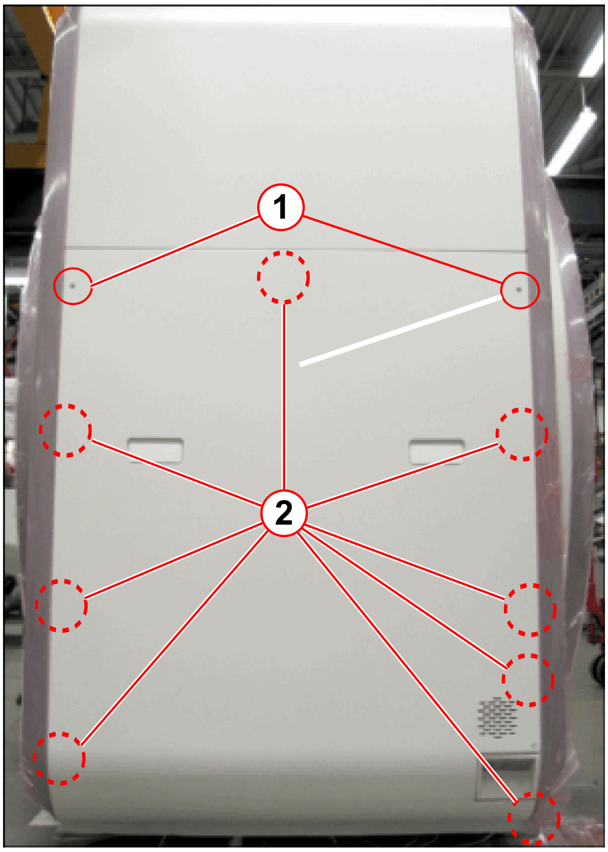
Fig. 9: Snap lock pin

Fig. 10: Snap lock bushing



> Snap locks.
> Do not use a screwdriver to remove the cover. Otherwise, the cover may be damaged.

Fig. 11: Lower service cover

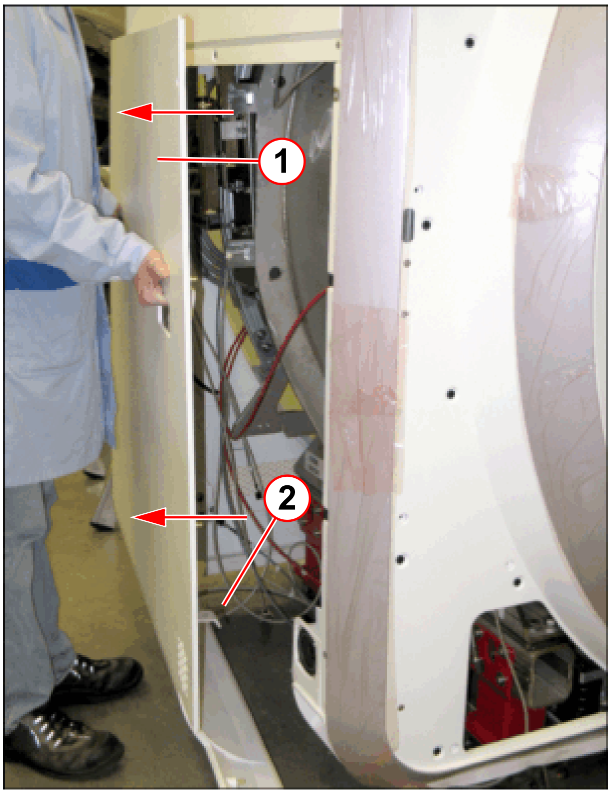


(1) Screws
(2) Snap lock positions

- Remove the screws (1).

Carefully pull out the lower service cover so that the snap locks are not damaged.

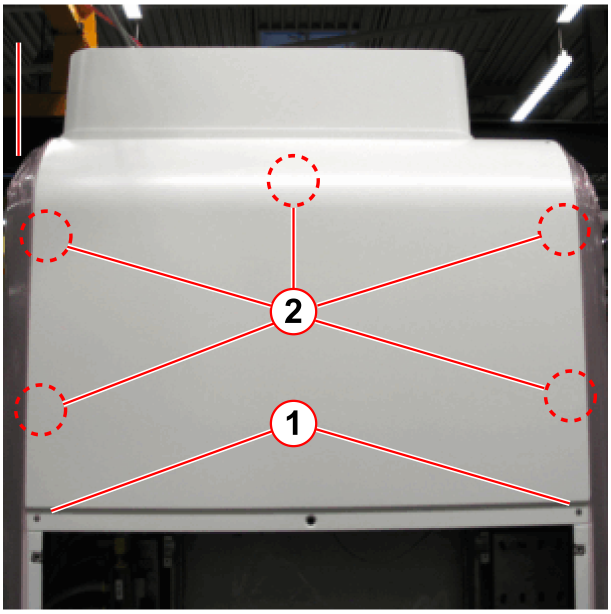
Fig. 12: Removed lower service cover



(1) Lower service cover
(2) Bracket

- Carefully pull out the lower service-cover (plugged with snap locks).

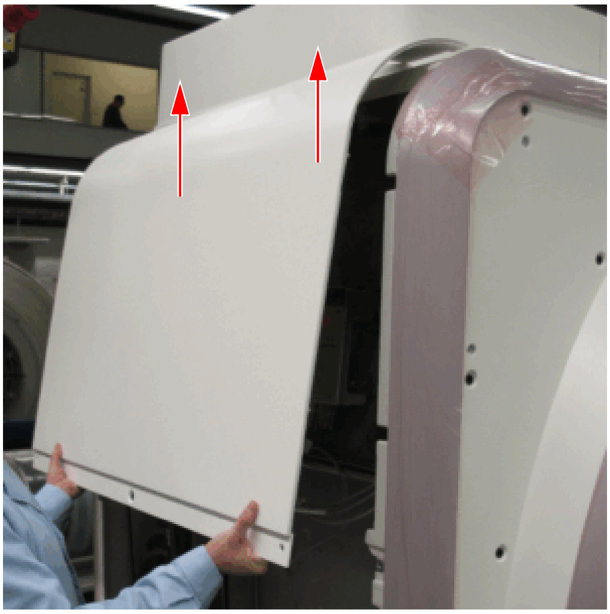
Fig. 13: Upper service cover



(1) Screws
(2) Position of snap locks

- Remove the screws (1).

Fig. 14: Upper service cover

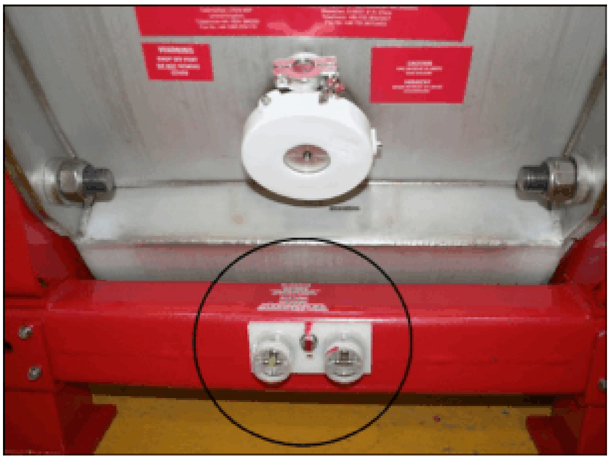


- Pull out and slide up to remove the upper service cover.
- This cover is removed for access to the **MSUP.**

Shock indicators

Every magnet shipped is fitted with 2 shock indicators. The indicators activate at impacts of **2g** and **6g**.

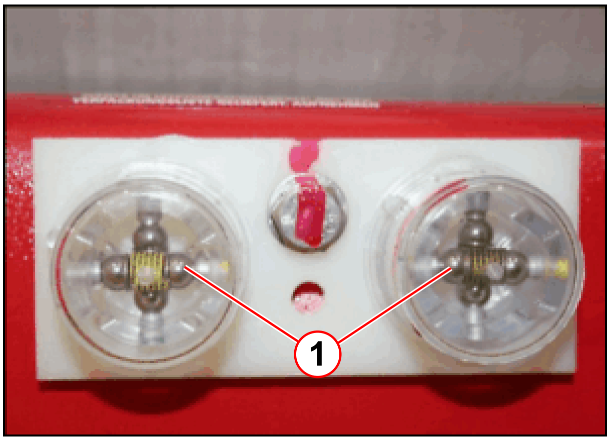
Fig. 15: Shock indicator position (right-hand transport box section)



(1) Shock indicator

- Remove the cover to access the shock indicator.
- **Check** the **status** of the **shock indicators**.

Fig. 16: Shock indicators (normal)



- The figure to the left shows how the **shock indicators** should look with everything **normal**.
- If the balls have been dislodged, **Siemens HSC** must be informed immediately, before the magnet is moved any further.

  (→ Magnet HSC, mailto:magnet.wscshsc.healthcare@siemens.com)

1.1.1.2 Equipment and tools required

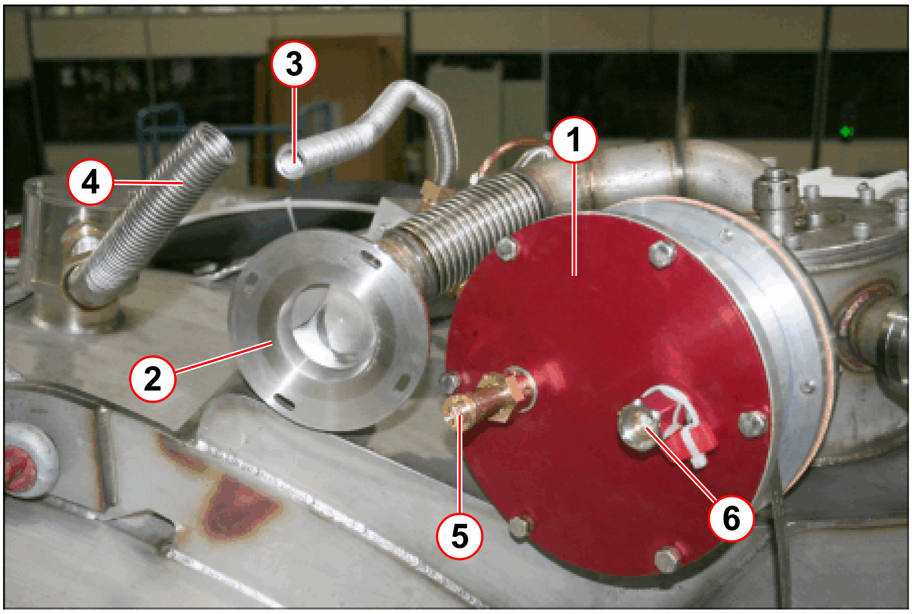
- Magnet **service syphon kit** (part no **10126651 (Revision 01)**). (→ Fig. 34 Page 31)
- **External 36V PSU kit** (service tool - **part number 08396371**) to provide power to the MSUP.
- **External 36V PSU MSUP cable** (service tool - **part number 10612362**).
- Dewar syphon.
- Dewar neck adaptor, clamps, seals, and hose.
- Heat gun.
- Paper towelling.
- Helium gas supply (at least 99.9% pure).
- Pressure regulator 200 bar/ 0 - 1.5 bar.
- Protective clothing: gloves, safety glasses, face shield, lab coat, safety shoes.
- Replacement copper seal for syphon port on the turret (part number **03108094**).
- Gas bottle wrench.
- A room or personal O_2 Monitor.
- Edwards hand-held **leak detector** - (part number **10614850**).
- Suitable service platform or step ladder.
- Service tool kit.
- 4mm Allen key (removing the magnet covers).
- Laptop computer and RS232 cable, (if available)

1.1.1.3 Check the Transportation kit

For all modes of transport (air or sea), the **quench valve** must be **isolated** from changes in **atmospheric** pressure.

A **red transport plate** is fitted to the front flange of the quench valve. In the event of a error, the **secondary** vent is via the **13.0psi (G) relief valve.**

Fig. 17: Transport configuration - OR98/OR99/OR103

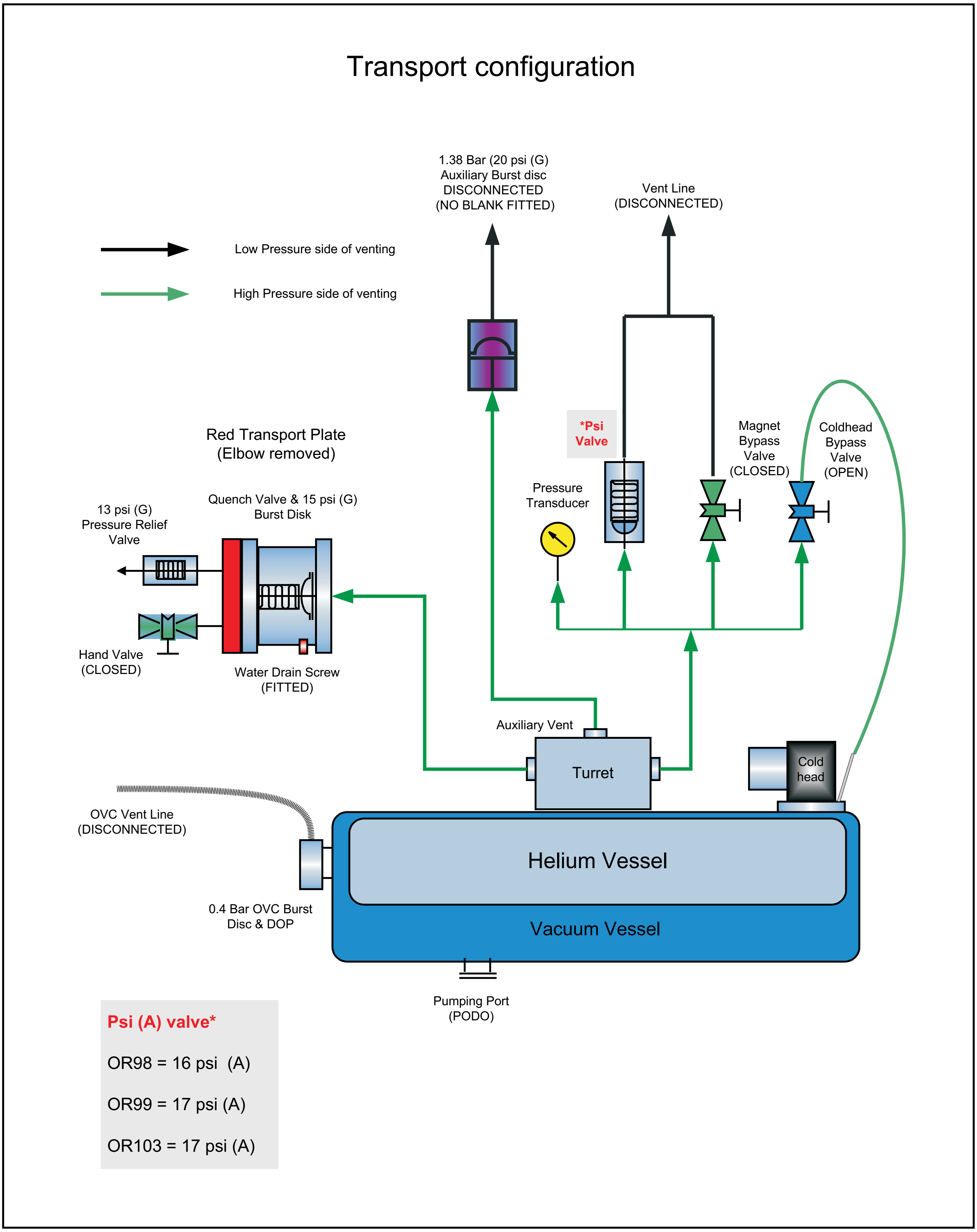


(1) Transport plate
(2) Auxiliary burst disk (open)
(3) Vent valve (open)
(4) OVC burst disc vent (open)
(5) 13 psi (G) valve
(6) Hand valve

NOTE: The figure above shows the venting configuration without the protective wrapping.

The magnet configuration must be checked on arrival. Any deviations from the transport configuration must be reported to the Magnet HSC. (➔ Magnet HSC, mailto:magnet.wscshsc.healthcare@siemens.com)

Fig. 18: OR98/OR99/OR103 Transport Venting Configuration



Tab. 4 OR98/ OR99 / OR103 checklist for venting configuration.

	Checklist of transport venting	Yes/No
1	Check **magnet gauge pressure.** **16 psi (A) OR98** - Expected pressure is **0.7 - 2.3 psi (G)**. **17 psi (A) OR99 / OR103** - Expected pressure is **1.7 - 3.3 psi (G)** Pressure increases, depending on the altitude. Refer to (➔ Fig. 22 Page 21), (➔ Fig. 23 Page 21)	
2	Red **transport plate** fitted to quench valve	
3	**Hand valve** on red transport plate **closed**	
4	**Magnet** bypass valve **closed** and labeled	
5	**Low** pressure vent line **disconnected**	
6	**OVC** vent line **disconnected**	
7	**Auxiliary** burst disk line **disconnected and open**	
8	**Visual check - auxiliary burst disk** intact	
9	**Cold head** bypass valve **open and labeled**	
10	**Metal drain screw** fitted to underside of **Quench Valve**	
11	Venting **connection kit** parts (attached to magnet)	
12	**Warning labels** and 'Flash card' fitted	
13	Fit **36 Volt PSU** to the **MSUP** and **check helium level**	
14	**Check** magnet **temperature LEDs**	

WARNING

Low or Zero magnet pressure.

Venting open to atmosphere will cause air ingress and icing of the helium vessel.

» **Check that the magnet bypass valve is CLOSED.**

» **Check that the auxiliary burst disk is INTACT.**

WARNING

Risk of air ingress

The auxiliary burst disk is broken.

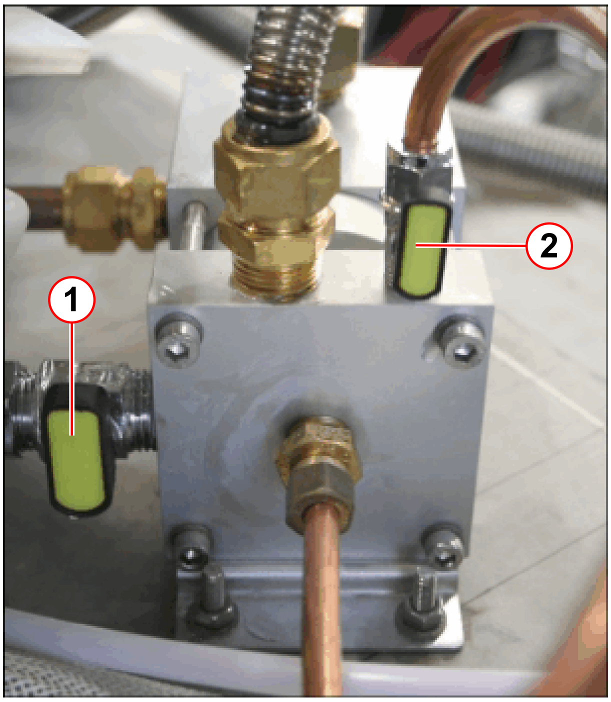
- **IMMEDIATELY fit the red blank from the customer spare parts box.**
- **Contact Magnet HSC immediately! (➔ Magnet HSC, mailto:magnet.wscshsc.healthcare@siemens.com)**

1.1.1.4 Check the bypass valve positions

The transportation positions of the venting bypass valves must be set as shown in the picture below.

These valve settings apply for both air and sea shipment.

Fig. 19: Venting valve positions



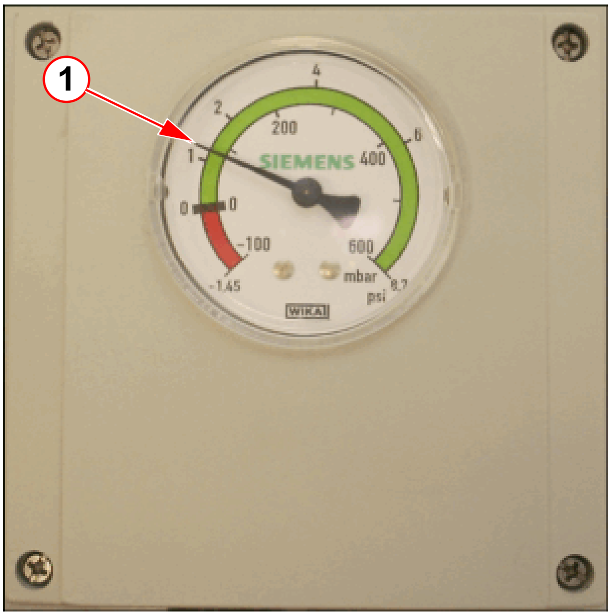
(1) Magnet bypass valve CLOSED
(2) Cold head bypass valve OPEN

- The **magnet bypass valve** (1) must be **CLOSED**.
- The **cold head bypass valve** (2) must be **OPEN**.

1.1.1.5 Check the magnet pressure

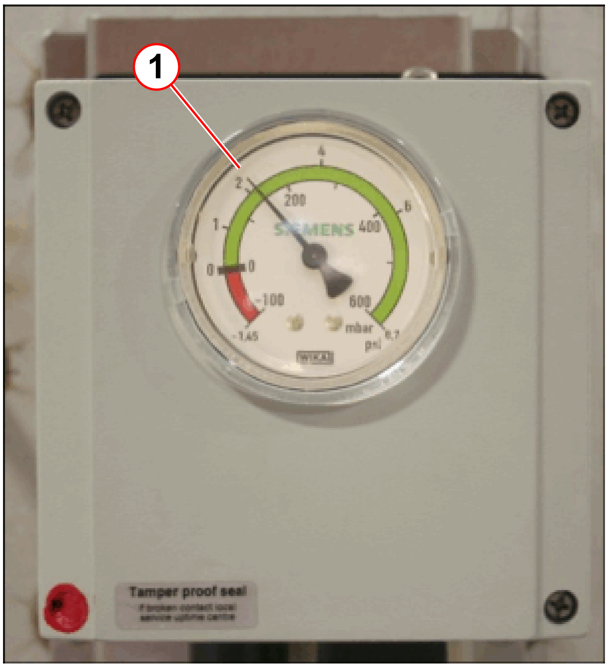
16 psi (A) valve fitted	17 psi (A) valve fitted

Fig. 20: OR98 - Pressure gauge



(1) Expected gauge pressure 0.7 - 2.3 psi (G)

Fig. 21: OR99/OR103 - Pressure gauge (expected pressure)



(1) Expected gauge pressure 1.7 - 3.3 psi (G)

- The mechanical gauge displays the relative magnet pressure (= gauge pressure) to atmosphere in psi (G).

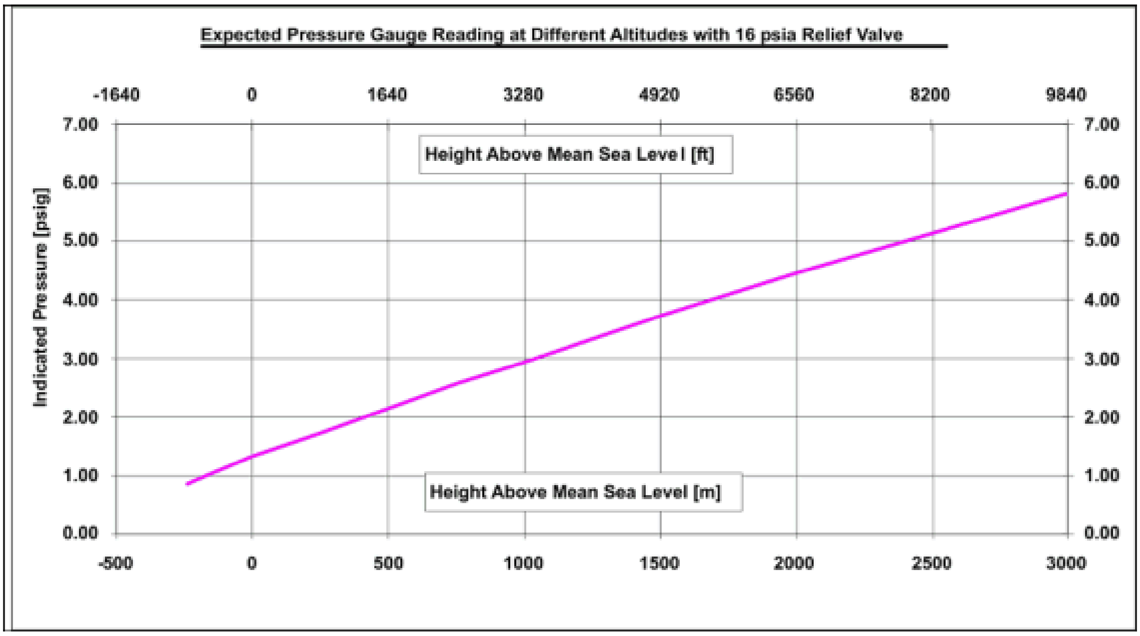
Tab. 5 Magnet pressure OR98/OR99/OR103

	OR98	OR99/OR103
Magnet pressure	16 psi (A) 1.3 psi (G)	17 psi (A) 2.1 psi (G)
Typical range at **sea level** pressure gauge reading (atmospheric variation)	0.7 - 2.3 psi (G)	1.7 - 3.3 psi (G)
Magnets installed at 2000 m (example)	4.5 psi (G)	5.2 psi (G)

The system may be left venting safely through the **vent valve** until installation.

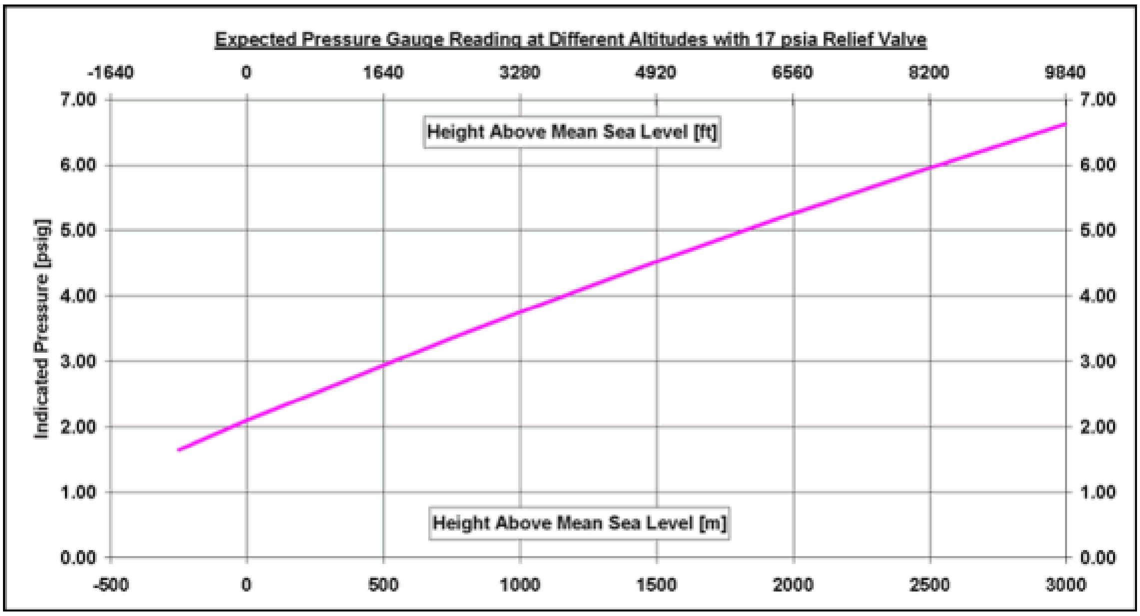
Altitude effect - 16 psi (A)

Fig. 22: Effect of a change in altitude on gauge pressure (16 psi (A))



Altitude effect - 17 psi (A)

Fig. 23: Effect of a change in altitude on gauge pressure (17psi (A))

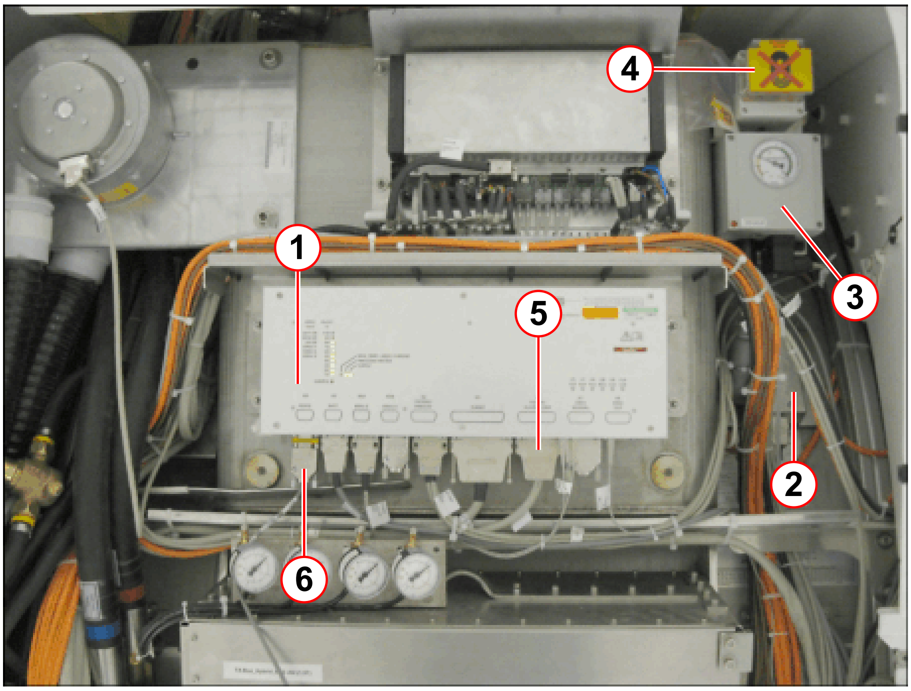


1.1.1.6 Connecting power to the MSUP

When the MSUP is powered, measurements can be taken using a Laptop.

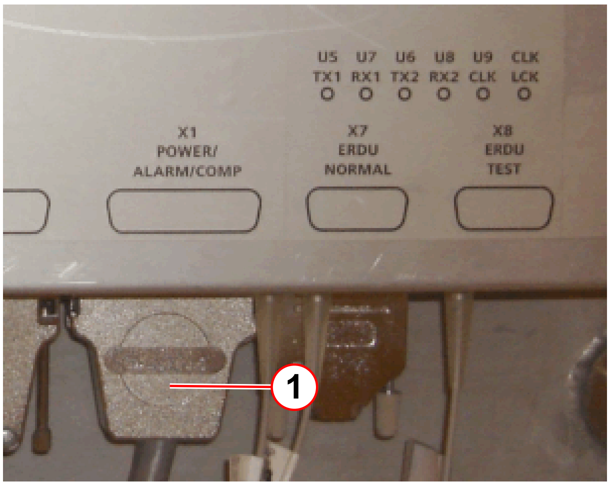
If a Laptop computer is available, connector to X6 on the MSUP via RS232 cable. (→ Using the Laptop to check magnet parameters / Page 77). For more information, refer to the(→ In-Transit Troubleshooting Guide / Page 74).

Fig. 24: MSUP 2306 position (static systems)



(1) MSUP
(2) ERDU battery
(3) Pressure gauge
(4) Magnet Stop Button (ERDU)
(5) X1 (POWER/ALARM/COMP)
(6) X6 RS232 (laptop)

Fig. 25: Connect the 36V power supply at X1

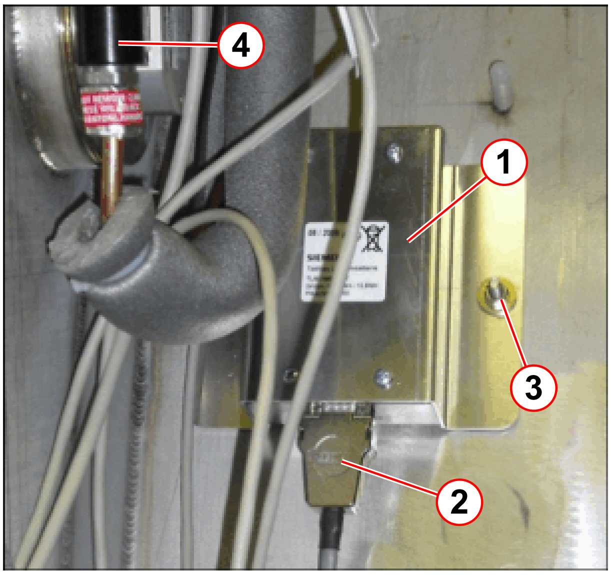


(1) Connector X1

To enable **magnet readings to be taken**, the **MSUP** must be **connected**, via an external 36V power supply.

- Connect the **external PSU kit** (service tool - **part number 10614850**) to provide power to the MSUP.
- Connect the **mains lead** between the **power supply** and **local power source**.
- Connect the **adaptor lead** between **X1 MSUP** (D type connector) and the **power supply.**
- **Turn ON** the **36 V** local **power** source.

Fig. 26: ERDU battery box location

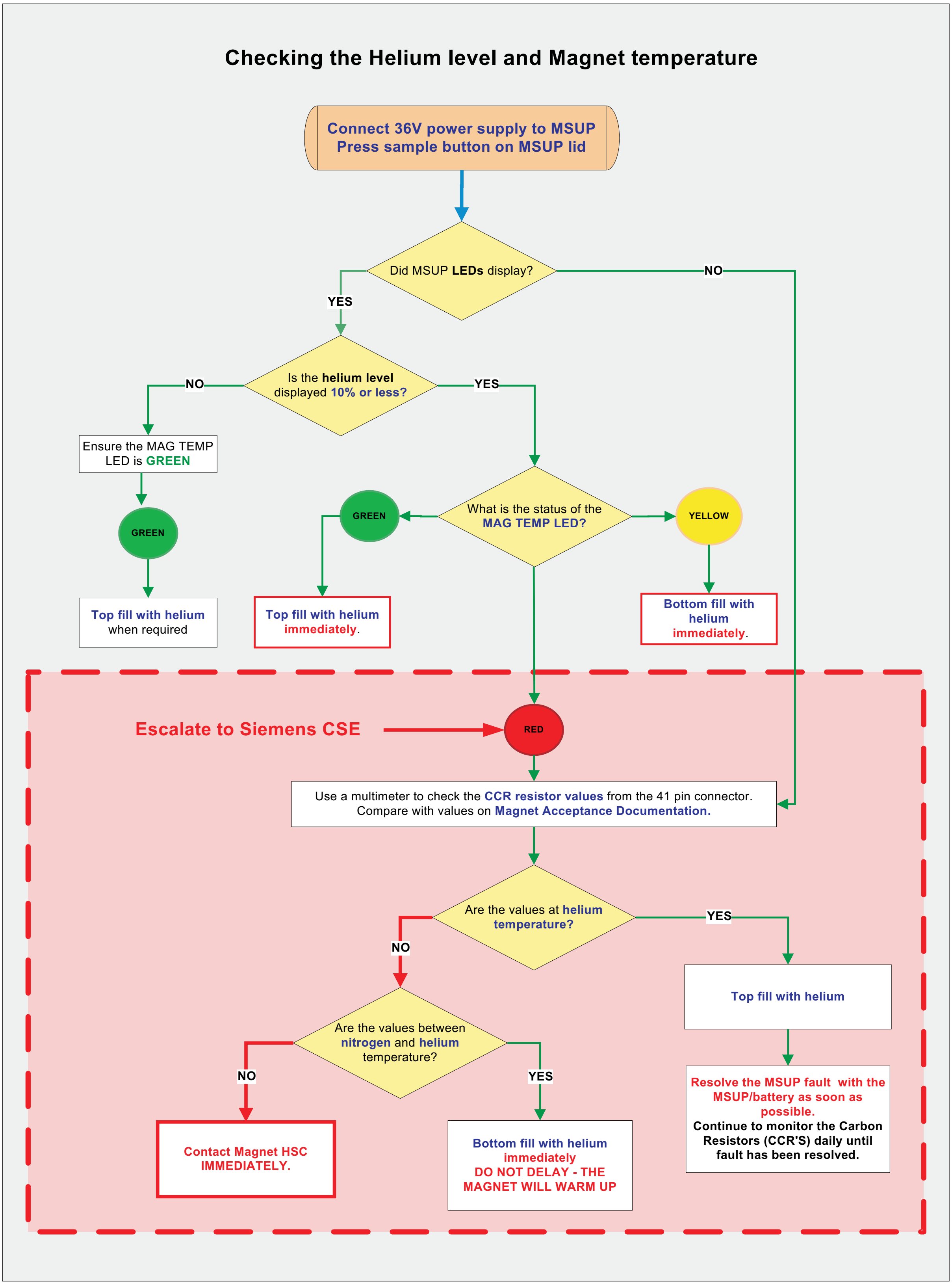


(1) ERDU battery box
(2) ERDU battery cable
(3) Nut and washer (2 off)
(4) Pressure gauge

An external **24 V Lithium non-rechargeable battery** is mounted on the magnet and provides a backup to the ERDU function.

During magnet transportation, and also **In-Transit filling,** the standard 24 V Lithium Battery must stay **disconnected** to avoid discharge.

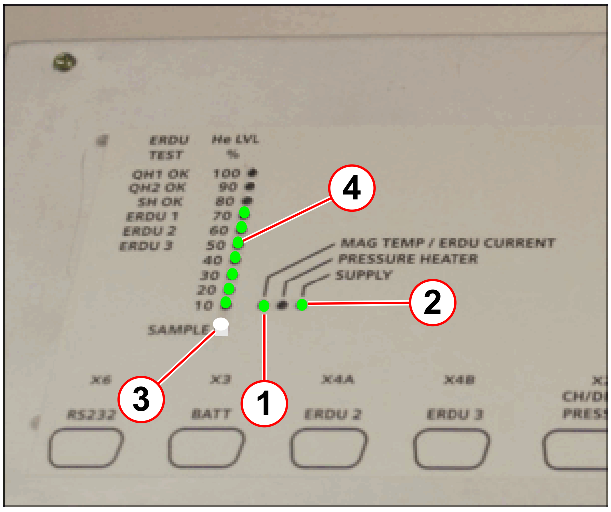
Fig. 27: OR98 / OR99 / OR103 He level and Magnet temperature check



1.1.2 Check the magnet temperature status

> The **MAG TEMP** status is indcated by the a TRI COLOUR LED. The LED will change from Red, Yellow or Green. (→ 1/Fig. 28 Page 25)

Fig. 28: Check the magnet temperature



(1) MAG TEMP/ERDU CURRENT
(2) SUPPLY
(3) SAMPLE BUTTON
(4) He LVL %

- At the MSUP, check the **colour** of the MSUP **MAG TEMP** LED. (→ Tab. 6 Page 25)

Tab. 6 MSUP 2306 - Magnet temperature indicators

Temperature	LED status	Action	
Room to nitrogen (77K)	RED	Check the CCR temperatures	Contact **magnet HSC IMMEDIATELY**
Nitrogen to helium	YELLOW	Refill **immediately**	**Bottom** fill
Helium temperature normal (4.2K)	GREEN	Refill as required	**Top** fill

- If the **LED** status is **RED** the magnet **CCR temperatures must be checked.**
  » Refer to the **Troubleshooting Guide**.

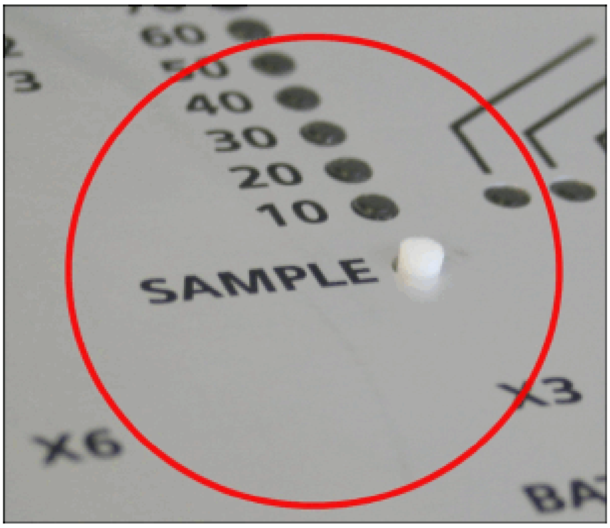
1.1.3 Perform a helium sample at the MSUP 2306

Check the **helium level** by pressing the MSUP **SAMPLE** switch.

Transit mode.

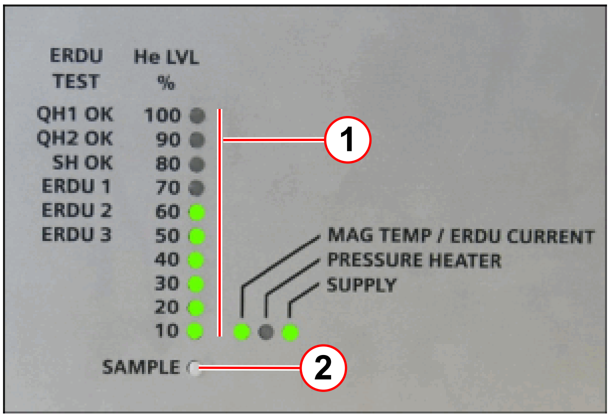
The liquid helium level can only be checked when the external 36 V power supply is connected.

Fig. 29: MSUP 2306 - depress SAMPLE to activate



- Locate the **SAMPLE** switch on the MSUP lid.
- Depress the sample switch for 2 s, and release.

Fig. 30: MSUP 2306 - Helium level sample (transit mode)



(1) Helium level % scale
(2) Sample switch

- The **green LEDs** illuminate, indicating the **actual liquid helium level.**
- Wait until the **sample is completed** (lights remain **static**).

If no LED lights appear, the MSUP may be faulty. Correct the error as soon as possible.

Check cable and plug connections to the MSUP, and 41 pin connector on the turret.

Escalate to Magnet HSC. (→ Magnet HSC, mailto:magnet.wscshsc.healthcare@siemens.com)

1.1.4 Determine which type of helium fill is required

- **Top** fill refer to (→ Assemble the service syphon - top fill / Page 44).

- **Bottom** fill refer to (→ Bottom filling the magnet / Page 59).

1.1.5 Third party helium filling

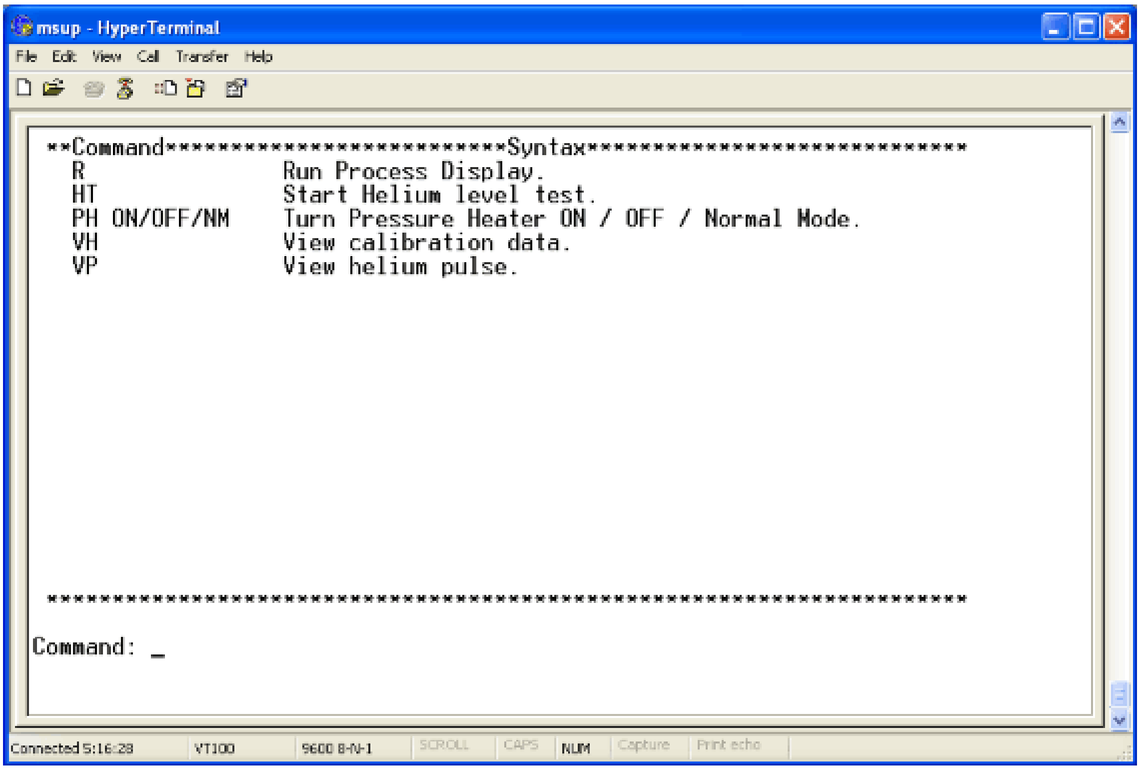
1.1.5.1 Perform a helium sample using a laptop PC

A Siemens password is required to access the full MSUP command screens.
Only Siemens CSE's have access to the password.

Third party companies have limited access to the MSUP command screens.

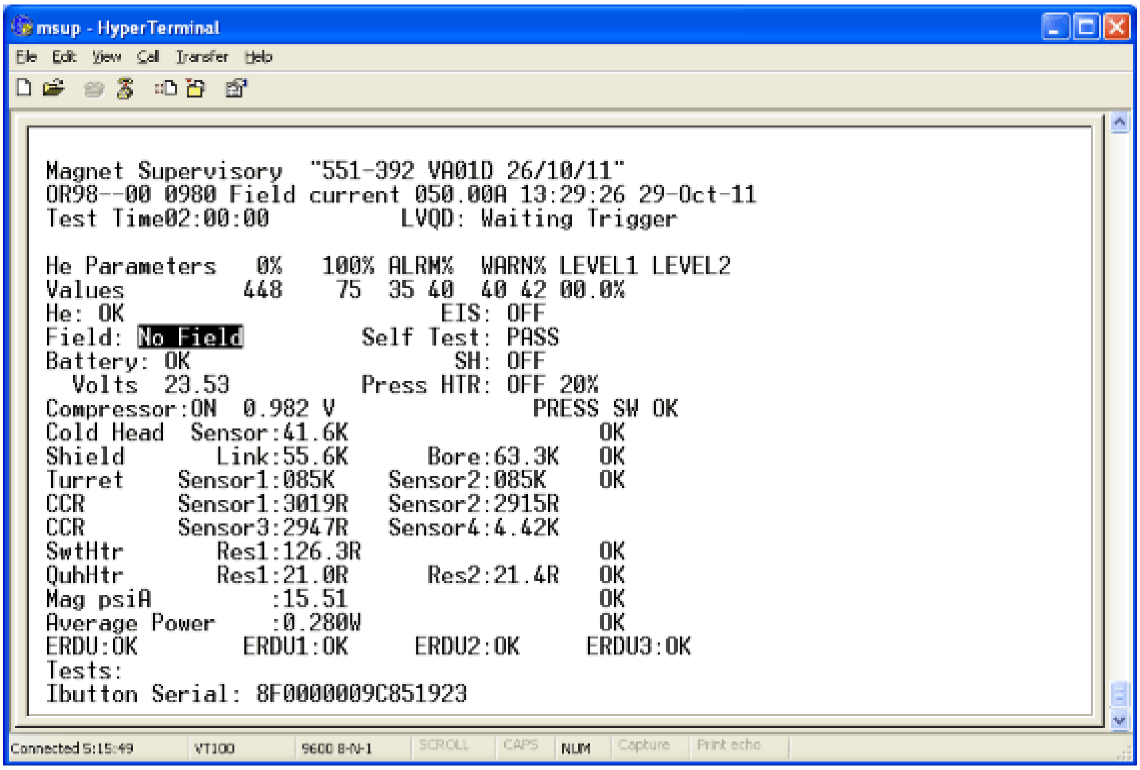
- Turn on the **laptop** and open the **Hyper Terminal application**.
  - Set bits per second to 9600.
  - Set data bits to 8.
  - Set parity to 1.
  - Set flow control to none.
  - Set emulation to VT100 (File: Properties: Settings).
- **Connect** the **laptop** to connector **X6** on the MSUP via **RS232** cable.
- **Press <ESC>** to display the **MSUP Option screen** (for third party companies).

Fig. 31: Option screen (for 3rd Party companies)



- Press **<R>** enter to display the **MSUP Process display screen**

Fig. 32: MSUP Process display screen (Example)



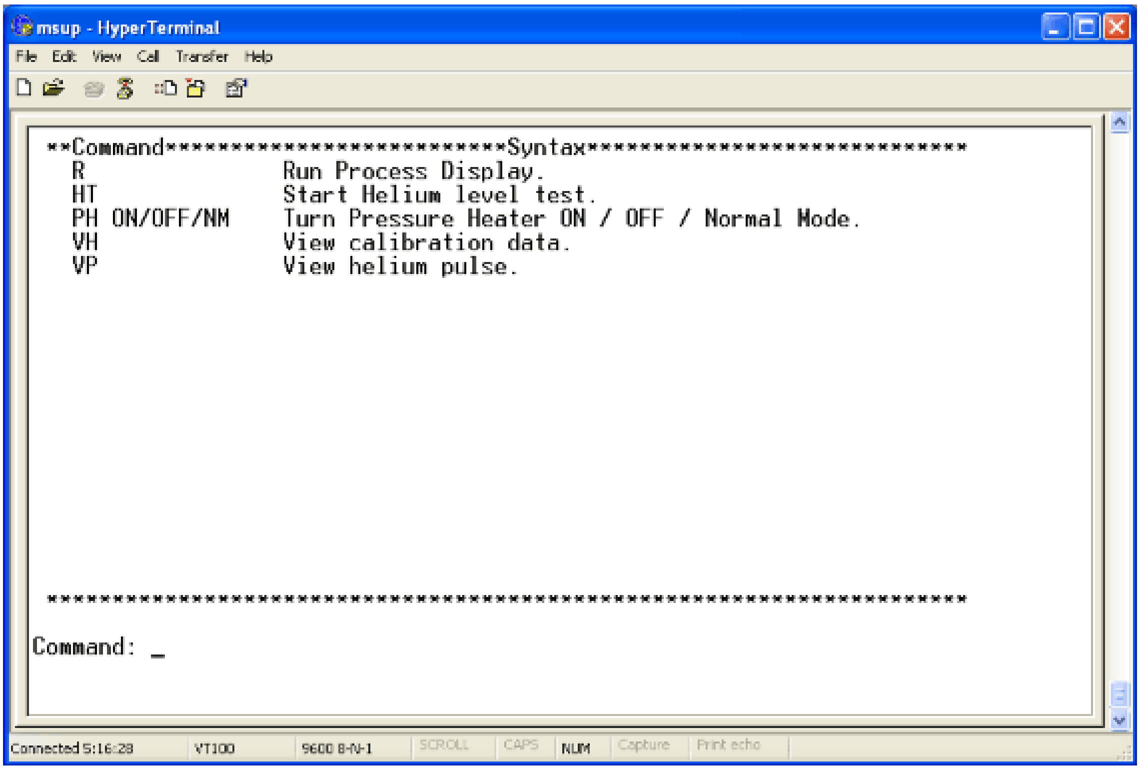
```
Magnet Supervisory  "551-392 VA01D 26/10/11"
OR98--00 0980 Field current 050.00A 13:29:26 29-Oct-11
Test Time02:00:00          LVQD: Waiting Trigger

He Parameters   0%    100% ALRM%  WARN% LEVEL1 LEVEL2
Values         448     75  35 40  40 42 00.0%
He: OK                         EIS: OFF
Field: No Field          Self Test: PASS
Battery: OK                     SH: OFF
  Volts  23.53           Press HTR: OFF 20%
Compressor:ON  0.982 V                PRESS SW OK
Cold Head   Sensor:41.6K                   OK
Shield        Link:55.6K      Bore:63.3K   OK
Turret     Sensor1:085K    Sensor2:085K    OK
CCR        Sensor1:3019R   Sensor2:2915R
CCR        Sensor3:2947R   Sensor4:4.42K
SwtHtr        Res1:126.3R                  OK
QuhHtr        Res1:21.0R      Res2:21.4R   OK
Mag psiA          :15.51                   OK
Average Power     :0.280W                  OK
ERDU:OK        ERDU1:OK     ERDU2:OK     ERDU3:OK
Tests:
Ibutton Serial: 8F0000009C851923
```

Take a new helium sample

- Press **<ESC>** to display the **MSUP Option screen (for third party companies).**

Fig. 33: Option screen (for 3rd Party companies)



- Press **HT <ENTER>.** Wait until the sample is **completed**.
- Press **R <ENTER>** to display the **new helium value** on the **MSUP Process display screen.**

Tab. 7 MSUP Commands

MSUP RS232 command, followed by **<ENTER>**	System Operation
HT	A new helium sample
PH ON	Turns **on** the pressure heater
PH OFF	Turns **off** the pressure heater
PH NM	Turns the pressure heater to the normal running mode
VH	Displays the helium calibration parameters
VP	Displays the helium pulse information
R	Displays the process image

1.1.6 MSUP power

With the 36 Volt providing power to the 2306 MSUP, the magnet pressure heaters will be activated when the magnet pressure drops to **below 0.2 psi (G).** This function will occur when the magnet LHe vessel is depressurized to allow fitment of the Service syphon.

To prevent the magnet pressure heater from activating:-

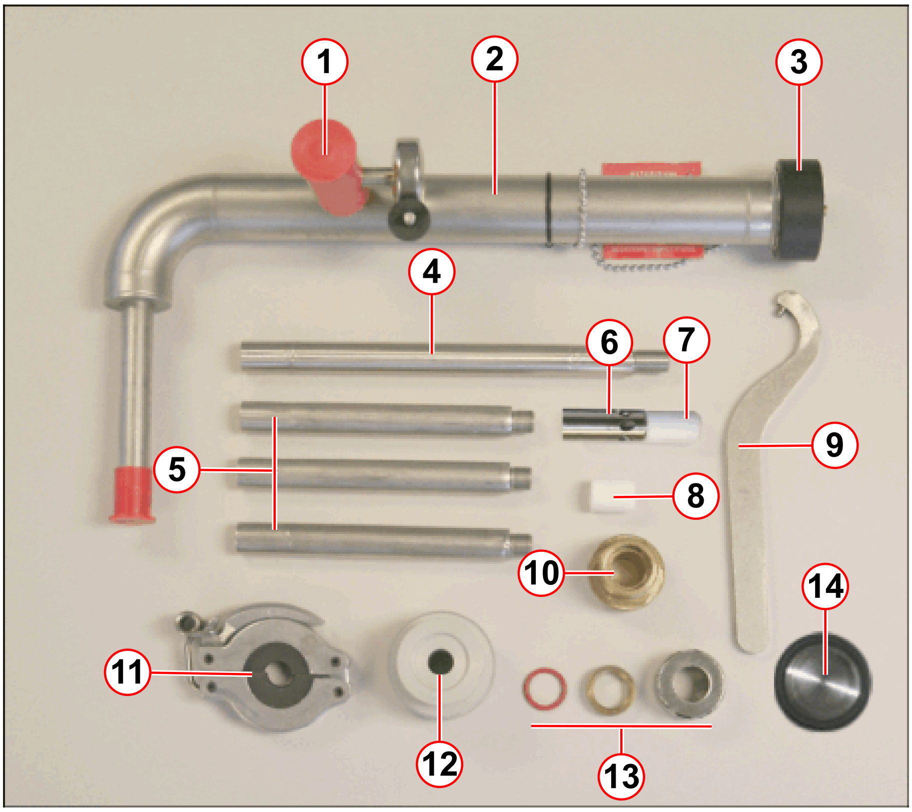
- **Disconnect** the 36 Volt power supply **during magnet de-pressurization.**
- **Re-connect** the 36 Volt power supply **when the LHe transfer has started.**

> Without the 36 Volt power to the MSUP the LHe level will not be monitored.

1.2 Filling the magnet during transportation

This procedure is restricted to Transportation filling only and must NEVER be used for filling magnets at field or when connected to an operational refrigeration system.

Fig. 34: Magnet service syphon kit



(1) Vacuum valve
(2) Service syphon assembly
(3) Syphon end cap
(4) 200 mm extension leg (1 off)
(5) 140 mm extension legs (3 off)
(6) 6 hole de-ice and helium fill diffuser tip (top fill)
(7) PTFE de-ice extension tip (top fill)
(8) 22mm PTFE seal (bottom fill)
(9) Syphon C-spanner/wrench
(10) M28 brass syphon adaptor bush (OR105, OR122)
(11) Syphon leg support clamp
(12) Nitrogen fill adaptor and O-ring seal
(13) Syphon ring nut, brass syphon flange, syphon O-ring seal
(14) NW25 flow restrictor

The magnet design does not include a permanently fitted syphon. A short **magnet service syphon** must be fitted when filling with liquid helium. The service syphon is tooling only and must be removed immediately after filling has been completed.

During filling, do not exceed 95%.

To recover a potentially warm magnet, (yellow MSUP LED) refer to magnet temperatures in the Bottom filling procedure (Bottom filling the magnet).

If a helium fill is showing poor fill efficiency, or no increase on the helium level meter, the transfer must be stopped and the syphon port inspected for ICE.

Only trained magnet CSEs are allowed to perform de-ice procedures.

Refer to Helium filling - Troubleshooting guide De-icing the syphon port.

1.2.1 Safety

Site preparation.

Ensure warning signs are correctly positioned.

Ensure there is a clear path to the nearest exit.

WARNING

Risk of asphyxiation.

Failure to observe the following safety warning may result in dizziness and loss of consciousness in personnel working with cryogen.

» **Oxygen monitors must be activated at all times.**

» **Helium gas will vent through the vent valve.**

» **In a storage facility, ensure there is adequate ventilation.**

CAUTION

Cold head failure.

The cold head seals can fail, and leak to atmoshphere.

» **Always ensure the cold head bypass valve is CLOSED during helium filling.**

CAUTION

Cryogenic liquids.

Failure to observe the following safety warning may result in cold burns.

» **Protective clothing, gloves and eye protection must be worn.**

CAUTION

Pressurized helium vessel.

Cold helium gas.

Failure to observe the following safety information may result in physical injury.

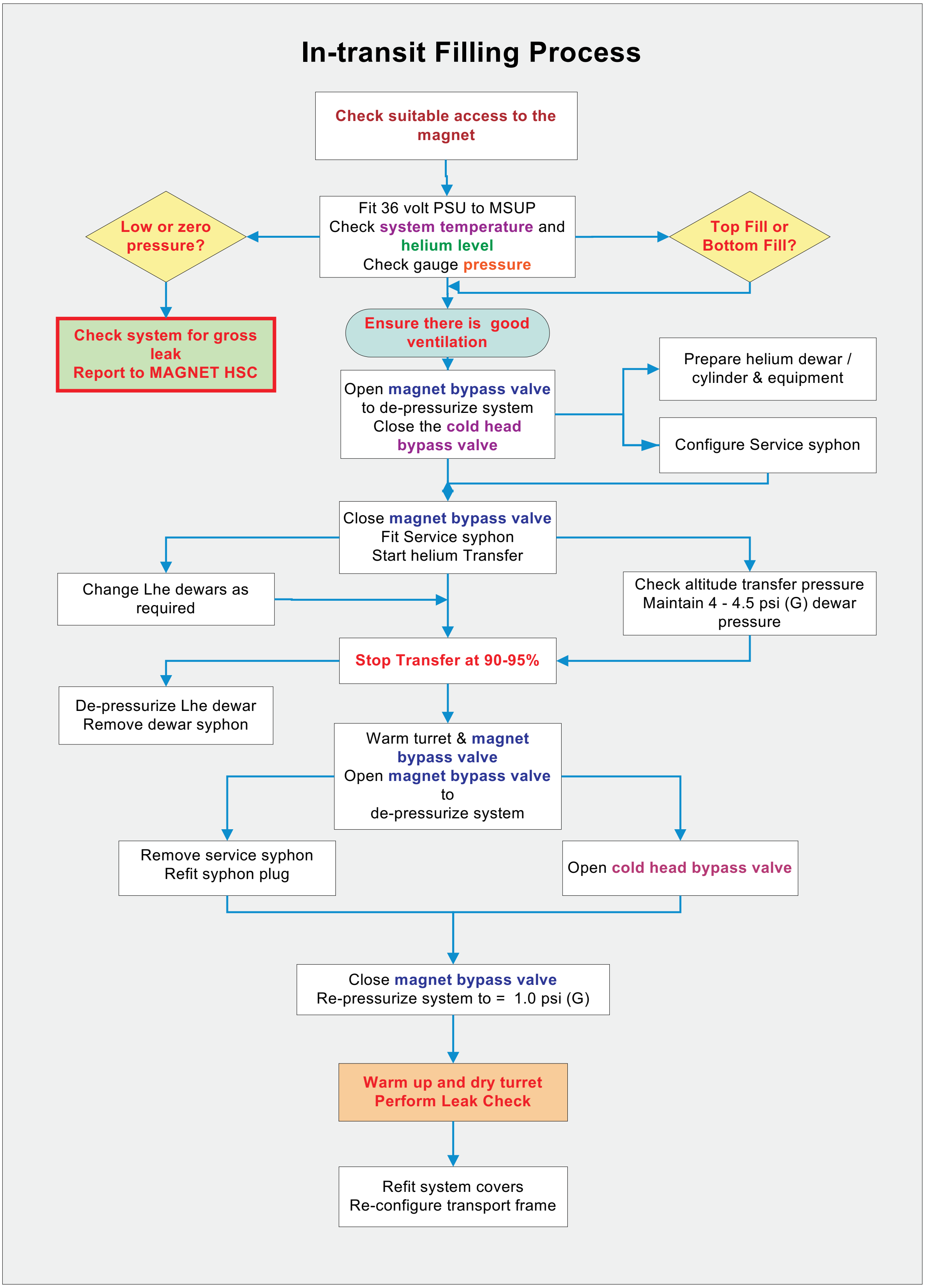
» **Do not disconnect any of the turret venting or fittings without first depressurizing the helium vessel.**

NOTICE

Mishandling of the dewar.

» **The dewar-relevant safety regulations of the manufacturer must be observed.**

Fig. 35: OR98 / OR99 / OR103 In-transit Filling Procedure



1.2.2 System preparation

Certain operations must be performed **without delay** to ensure efficient transfer:-

- Ensure that there is **open and clear access** to the magnet for **helium dewar(s).**
- Ensure that there is **open and clear access** to the **syphon**.
- **Position** a suitable **safety ladder** next to the magnet.
- Perform a helium sample using the **MSUP**. (➔ Perform a helium sample at the MSUP 2306 / Page 25)

1.2.2.1 Magnet helium volumes

Tab. 8 Magnet helium average volumes and usage

OR98			OR99 and OR103		
Probe %	**Litres**	**Top up helium required**	**Probe %**	**Litres**	**Top up helium required**
0	0	1500 ltrs	0	0	1350 ltrs
10	128	1350 ltrs	10	82	1200 ltrs
20	256		20	226	
30	384	1030 ltrs	30	388	925 ltrs
40	512		40	505	
50	640	750 ltrs	50	621	660 ltrs
60	768		60	733	
70	896	450 ltrs	70	851	400 ltrs
80	1024		80	970	
90	1152		90	1080	
100	1280		100	1190	
13 ltrs corresponds to 1%			12 ltrs corresponds to 1%		

1.2.2.2 Helium pressures at altitude

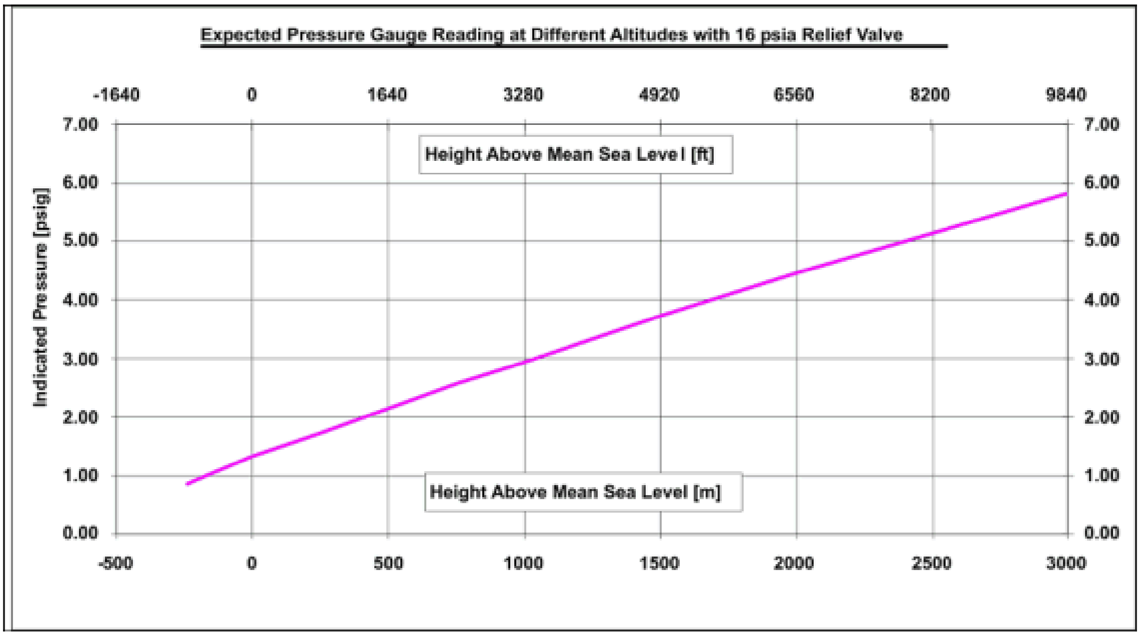
As a function of altitude, the graph(s) show the **gauge pressure** in the magnet.

The optimum pressure differential between magnet and dewar for a helium transfer is 4 psi (G).To calculate this:

- Determine the **magnet's altitude**.
- Determine the **magnet gauge pressure** from the graphs below.

Transfer pressure - 16 psi (A) valve

Fig. 36: Effect of a change in altitude on gauge pressure (16 psi (A))



Tab. 9 Dewar transfer pressures (typical)

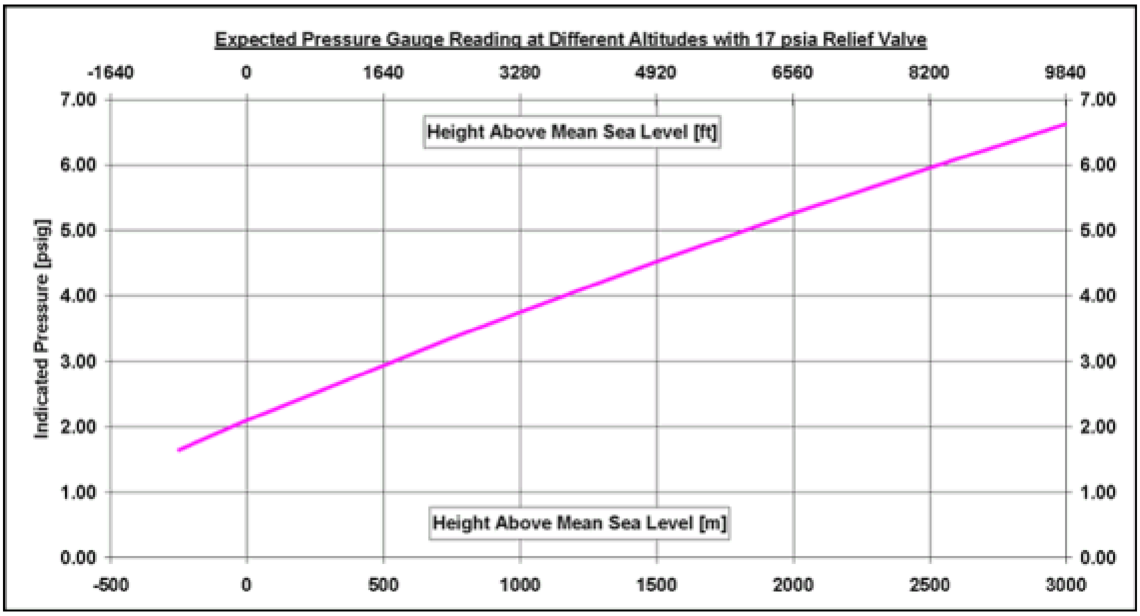
Altitude (M)	0	1000	2000	3000
Dewar pressure psi (G)	5	7.0	8.5	9.5

- **Add 4 psi** to this number to calculate the **required transfer pressure**.
  - » **Example of 16 psi (A) vent valve assembly**
    Altitude (1000m) = 3.0 psi (G) + 4 psi = 7.0 psi (G)

Maximum dewar pressure is 10.5 psi (G).

Transfer pressure - 17 psi (A) valve

Fig. 37: Effect of a change in altitude on gauge pressure (17psi (A))



Tab. 10 Dewar transfer pressures (typical)

Altitude (M)	0	1000	2000	3000
Dewar pressure psi (G)	6	7.75	9.25	10.5

- **Add 4 psi** to this number to calculate the **required transfer pressure**.
  - » **Example of 17 psi (A) vent valve assembly**
    Altitude (1000m) = 3.75 psi (G) + 4 psi = 7.75 psi (G)

Maximum dewar pressure is 10.5 psi (G).

1.2.3 Depressurize the helium vessel

- **Before** depressurizing the LHe vessel, **disconnect** the power to the 2306 MSUP.

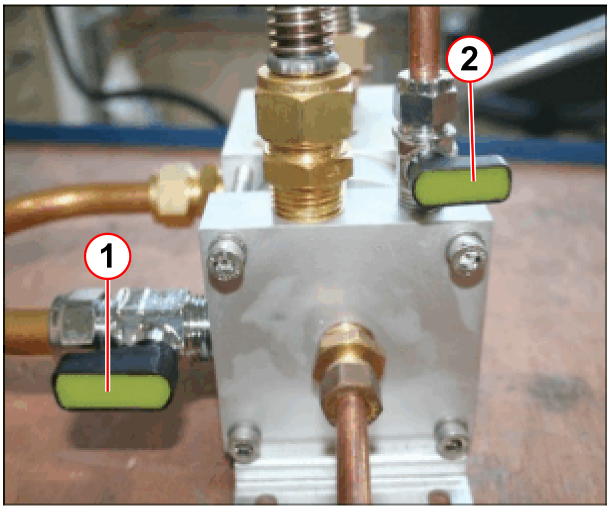
CAUTION

Risk of ice formation.

The magnet bypass valve is only OPENED for depressuization of the helium vessel. The bypass valve is CLOSED during normal operation.

» **To avoid air ingress, CLOSE the magnet bypass valve immediately after the helium vessel is depressurized.**

Fig. 38: Bypass valve positions for depressurization

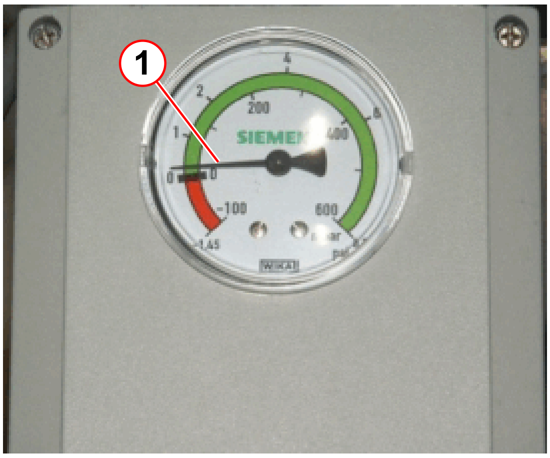


(1) Magnet bypass valve OPEN
(2) Cold head bypass valve CLOSED

- **Fully open** the **magnet bypass valve** to **depressurize** the helium vessel.
- Wait until **zero bar** is displayed on the pressure gauge.
- **Close** the magnet bypass valve to prevent **air ingress**.
- **Warm up** the vent valve assembly.
  » **Take care** not to damage any components when using a **heat gun**.

Depressurization could take up to 30 minutes or more.

Fig. 39: WIKKA Pressure switch - with a small positive pressure



- **Check** the helium vessel has a **small** remaining **positive pressure**.

During magnet de-pressurization, simultaneous operations for preparation can be performed:-

- Preparing the **dewar** and **dewar syphon**.
- Preparing the **service syphon** for fitment.

1.2.4 Prepare the helium dewar and equipment

1.2.4.1 Dewar preparation

This procedure describes the use of **top fill** helium dewars used in conjunction with the Siemens Magnet Technology dewar syphon.

The procedure covers all dewars that are externally pressurized with helium gas.

> Refer to the dewar manufacturer's documentation when using side fill and self-pressurizing dewars.
>
> Self pressurizing dewars only deliver 4 & 8 psi when the fill is performed at sea level. This may not be the most efficient pressure at higher altitudes. (→ Helium pressures at altitude / Page 35)

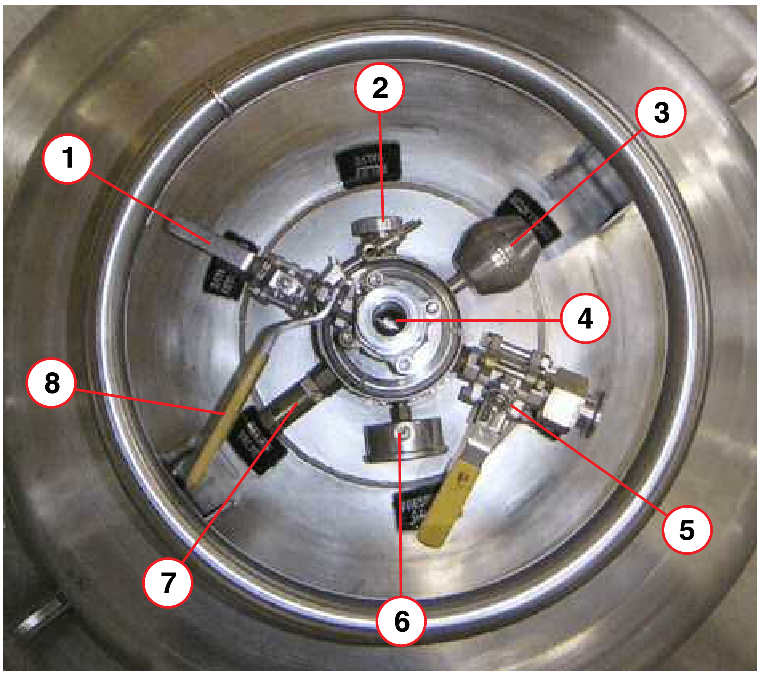
> Refer to the dewar manufacturer's safely guidelines and check the dewar valve functions before commencing the LHe fill.

- If the dewar is **highly pressurized**, (greater than 0.3 bar (4.4 psi (G)), release the pressure by slowly opening the **primary ball valve** (→ 5/Fig. 40 Page 40) to minimize the flash losses of helium.
  - This must be done in a **WELL VENTILATED AREA.**

> Check for the cause of the high pressure, i.e. was the low pressure relief valve closed?
>
> Inform the supplier if the cause cannot be found.

- Fit the **dewar syphon adaptor** (➔ 3/Fig. 42 Page 42) to the top of the dewar.

Fig. 40: Dewar Ports (typical)

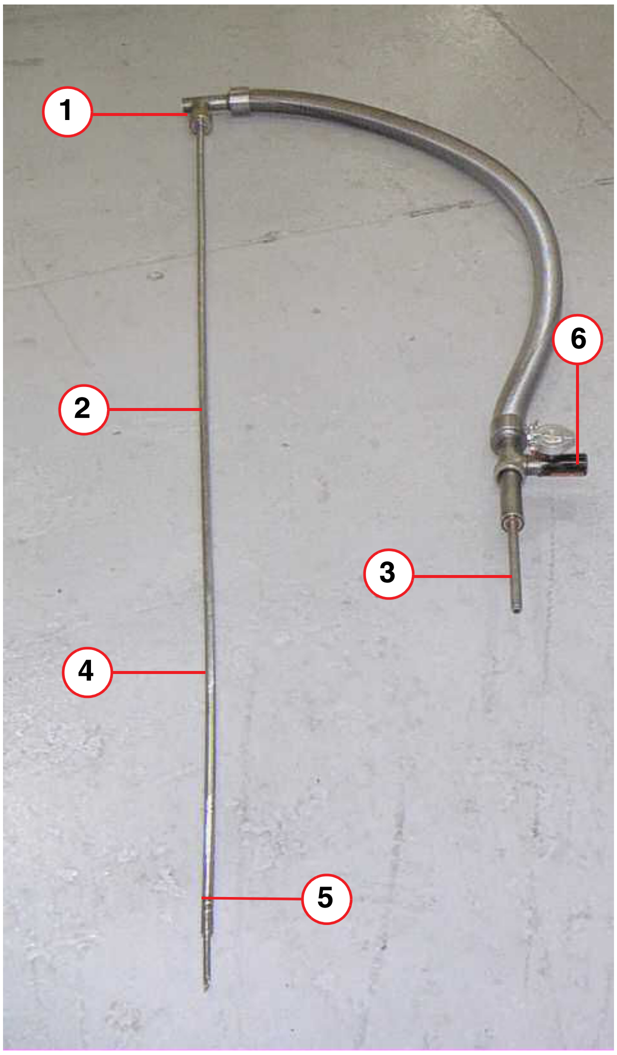


(1) Low pressure relief valve (shown open)
(2) Relief valve
(3) Anti-oscillator
(4) Syphon entry port (shown closed)
(5) Primary ball valve (shown closed)
(6) Pressure gauge
(7) Relief valve
(8) Syphon entry valve

- **Close** the **primary ball valve** (➔ 5/Fig. 40 Page 40) and **low pressure relief valve**. (➔ 1/Fig. 40 Page 40)

Dewar syphon preparation

Fig. 41: Dewar syphon



(1) Syphon flow valve
(2) Syphon leg
(3) Bayonet connection (male)
(4) Extension leg
(5) Sludge inhibitor
(6) Syphon vacuum valve

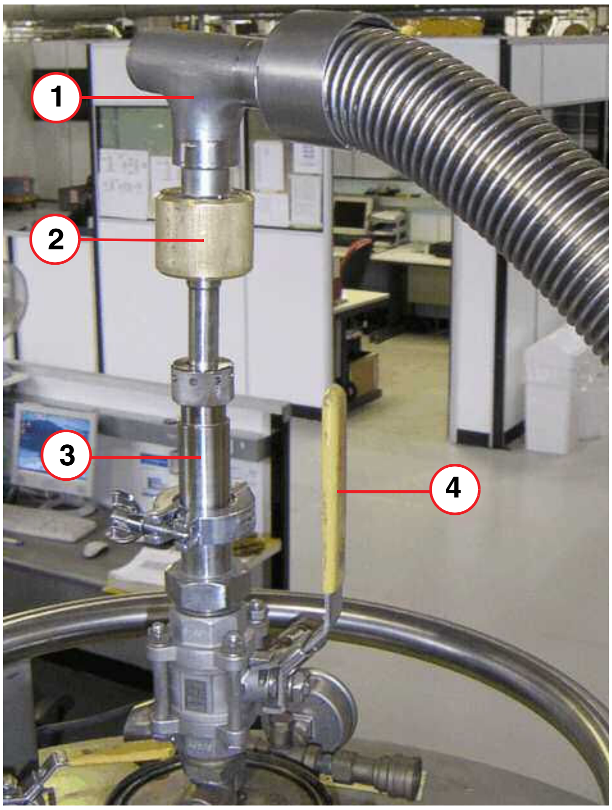
- Fit the extension leg (s) and sludge inhibitor, to the dewar **syphon leg**.

Do not open the syphon vacuum valve (6).

- Fit the **helium regulator** to the helium **gas bottle**.

Precool the dewar syphon

Fig. 42: Dewar syphon fitment



(1) Syphon
(2) Syphon flow valve
(3) Dewar syphon adaptor
(4) Syphon entry valve

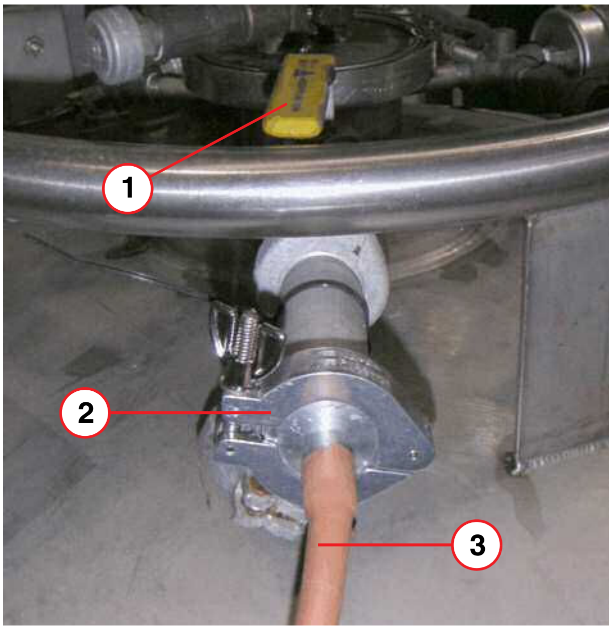
- Open the **syphon flow valve** (pos 2) two turns.
- Insert the syphon leg into the **dewar syphon adaptor** (pos 3), and tighten the ring nut and seal in order to create a gas seal that the syphon can slide through.

  (Do not fully tighten the ring nut at this stage).
- Open the **syphon entry valve** (pos 4), and slowly slide the syphon into the dewar. Gas will begin to flow through the syphon.

- Push the syphon **fully down** into the dewar, (very slowly to minimize pressure build-up and flash losses), until the sludge inhibitor touches the bottom.
- Do not allow high pressure to build in the dewar.
- Fully tighten the **ring nut** and **seal.**

- Move the dewar so that the syphon can be connected **without excessive bending**.

Fig. 43: Dewar gas connection



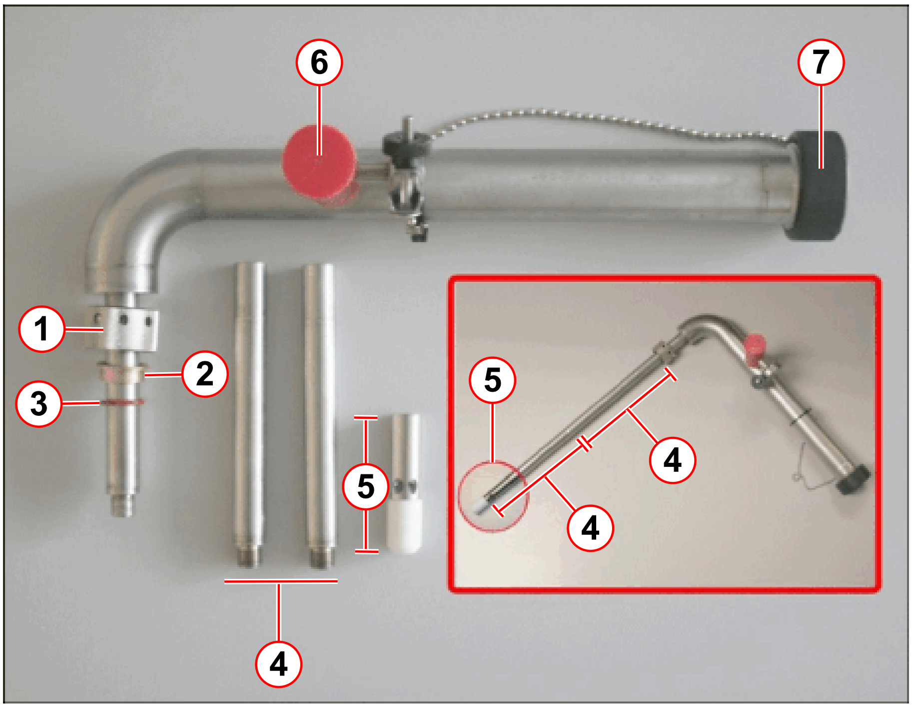
(1) Primary ball valve
(2) NW25 clamp and seal
(3) Gas supply line

- **Purge** the helium **gas supply line** of air.
- Connect it to the **primary ball valve** (1) on the dewar, using an **NW25** clamp and seal (pos 2).

- **Open** the **primary ball valve** (→ 5/Fig. 40 Page 40) and pressurize the dewar to **0.27 bar (4 psi (G))**.

1.2.5 Assemble the service syphon - top fill

Fig. 44: Magnet service syphon assembly - top filling



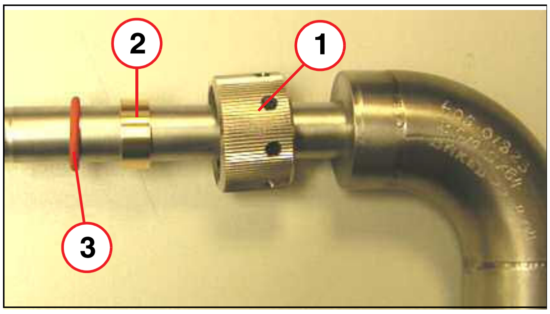
(1) Ring nut
(2) Brass spacer
(3) O-ring seal
(4) 140 mm syphon legs (2 off)
(5) De-ice and helium fill diffuser and PTFE tip
(6) Vacuum valve
(7) Syphon end cap

Tab. 11 OR98 / OR99 / OR103 TOP filling syphon extension legs

Magnet type	130* mm leg	60 mm De-ice & helium fill diffuser & PTFE tip
OR98, OR99 and OR103	2	1
The **syphon diffuser** should optimally be 50-80 mm above the syphon cone for **top filling**. **Note: *Effective extension leg length.** If damaged, the **PTFE diffuser tip** (➔ 7/Fig. 34 Page 31) must be replaced.		

- Assemble the **syphon**
  - Assemble the **diffuser**, bottom **extension** and support **clamp**.

Fig. 45: Assembling the Ring Nut Seal



(1) Ring nut
(2) Brass spacer
(3) Silicon O-ring

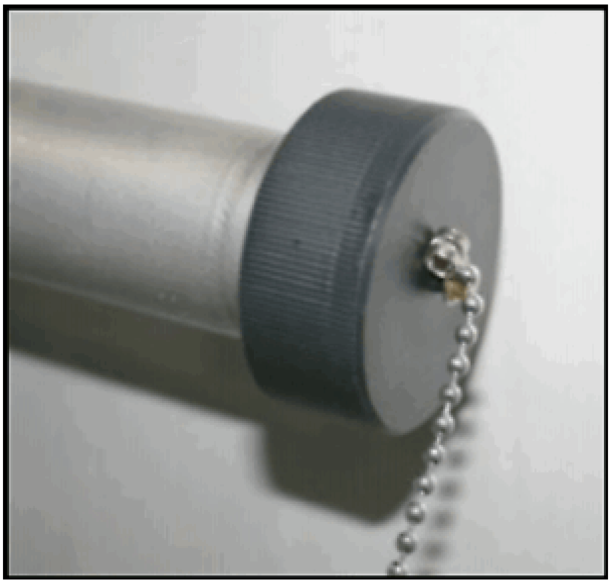
- Fit the ring nut, brass spacer, and the O-ring to the **service syphon leg.**

- Connect the **extension legs** and diffuser:
  - Screw the (2 off) **140 mm legs** together.[1]
  - Connect the leg to the **syphon**.
  - Fit the 60 mm **diffuser** and PTFE tip.

1.2.6 Fit the magnet service syphon

- **CLOSE** the magnet **bypass valve**.
- Warm and dry the **syphon blanking plug** using a heat gun to ease removal. (→ 1/Fig. 47 Page 46)

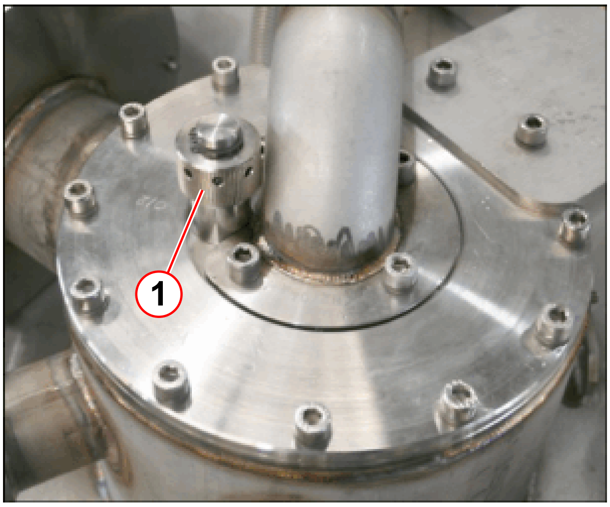
Fig. 46: Syphon end cap



- **Remove** the syphon **end cap.**

1. When connected, the effective length of the legs is 130 mm

Fig. 47: Fit/remove the syphon blanking plug



- **Remove** the **syphon blanking plug** using the syphon **'C' spanner**.

CAUTION

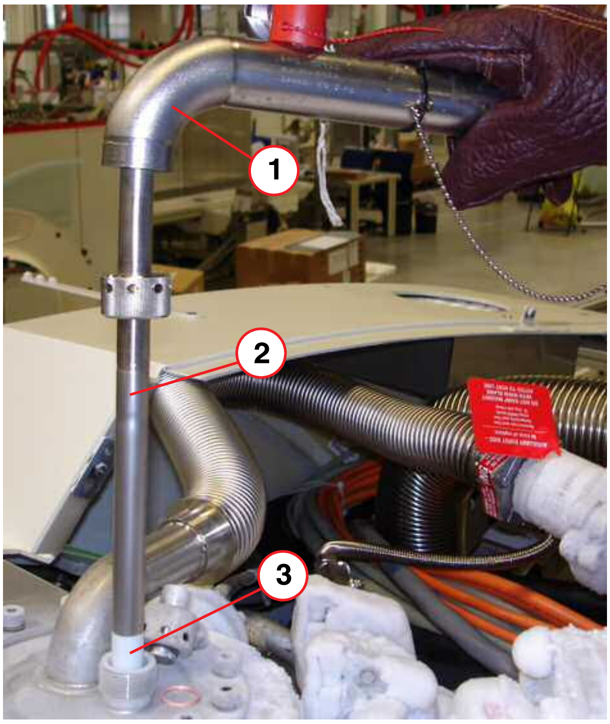
Risk of cold burns.

Cold gas will exhaust from the syphon port on opening.

» **Stand clear when opening.**

» **Wear protective gloves and safety glasses.**

Fig. 48: Helium syphon



(1) Service Syphon
(2) Two Extension Legs
(3) PTFE Syphon Diffuser

- **Insert** the service syphon into the port without delay.
- Slowly lower the service syphon fully into the magnet.
- Tighten the **syphon port ring nut** by hand, then apply a half turn using the syphon C-spanner to form a **gas tight seal**.

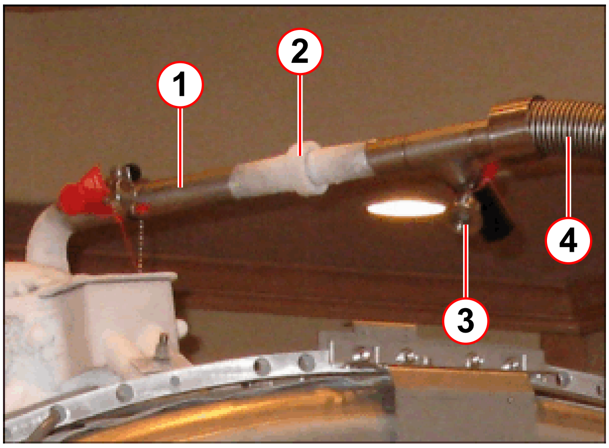
- **Re-fit** the syphon **end cap.** (→ Fig. 46 Page 45)

1.2.7 Liquid helium transfer - top fill

Before helium transfer, refer to (→ Helium pressures at altitude / Page 35)

- Check the **dewar** gas supply pressure is set at **0.27 bar (4 psi (G)).**
- **Purge** the **dewar syphon** with helium before coupling to the **service syphon**.

Fig. 49: Dewar and service syphon coupling



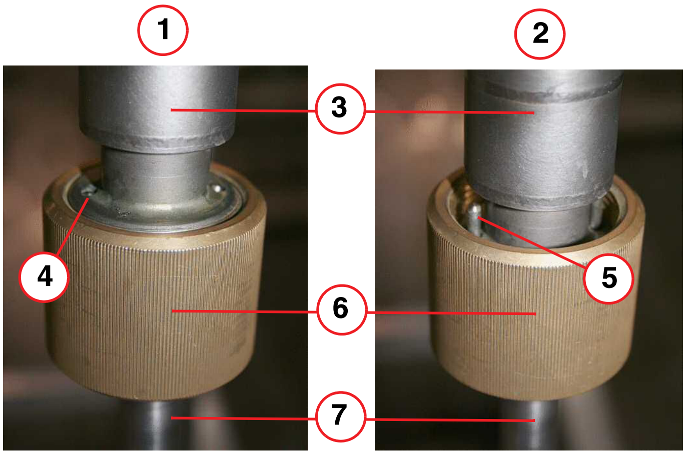
(1) Service syphon
(2) Lock nut
(3) Dewar syphon vacuum valve
(4) Dewar syphon

- Fit the dewar syphon into the service syphon and **tighten the locking nut.**

Frost or condensation on the body of either syphon is an indication of poor vacuum. Stop the transfer immediately - the syphon may need replacing.

Moderate frost on the syphon coupling (lock nut) (→ 2/Fig. 49 Page 47) is acceptable.

Fig. 50: Dewar syphon flow valves - open and closed positions



(1) FULLY OPEN position
(2) FULLY CLOSED position
(3) Stainless steel body
(4) Guide pin flush with body
(5) Guide pin
(6) Brass nut
(7) Syphon leg

Fig. 51: Helium filling

- Fully opening the **dewar syphon flow valve**.
  - » The flow valve is **fully open** when the two **guide pins** at the top of the brass nut are flush with the top of the stainless steel body (→ 4/Fig. 50 Page 48).
  - » From fully closed to fully open, is **4 turns**.
- Ensure the transfer dewar pressure is **4 - 4.5 psi (A)** at normal altitude.

1.2.7.1 Helium level check at the MSUP

To check the magnet helium levels **during filling**:-

- Re-connect **mains power** to the 36 Volt **power supply**.
  - **Leave** the MSUP **power connected** until the **end of the helium transfer.**
- **Monitor** the helium **level**:
  - at the MSUP LEDs, or laptop connection
- As soon as the **transfer has started** take a helium sample **every 5 minutes**.

Tab. 12 MSUP Helium display

MSUP LEDS	Helium level	Comments or actions
0 - 80% ON	As displayed	Continue filling.
90% ON	As displayed	**Stop** transfer.

MSUP LEDS	Helium level	Comments or actions
*90% **FLASHING** 10 - 80% ON	95% - 97%	STOP the transfer **IMMEDIATLEY!**
*100% **FLASHING** 10% - 90% ON	97% and over	Magnet is **OVERFILLED**.

* The LEDs return to ON after approx. 30 seconds.

To avoid overfilling, the helium transfer must be stopped as soon as the 90% LED illuminates.

With a laptop fitted the fill level can be stopped accurately at 95%.

- Repeat the measurement **every 5 minutes**. An increase in level should be seen within 20 minutes.
  - » If not, **stop the transfer** and investigate the reason for poor transfer efficiency.

Maintain a pressure of about 0.27 bar (4psi (G) at the helium dewar.

Too much or too little pressure will result in inefficiency.

At **4 psi (G)**, the transfer rate is approximately 10 liters per minute. **Do not leave the transfer unattended.** A 250 liter dewar will empty in 25 - 30 minutes.

If frost or condensation forms on the body of either syphon, it is an indication of poor vacuum.**Stop the transfer immediately - the syphon may need to be replaced.**

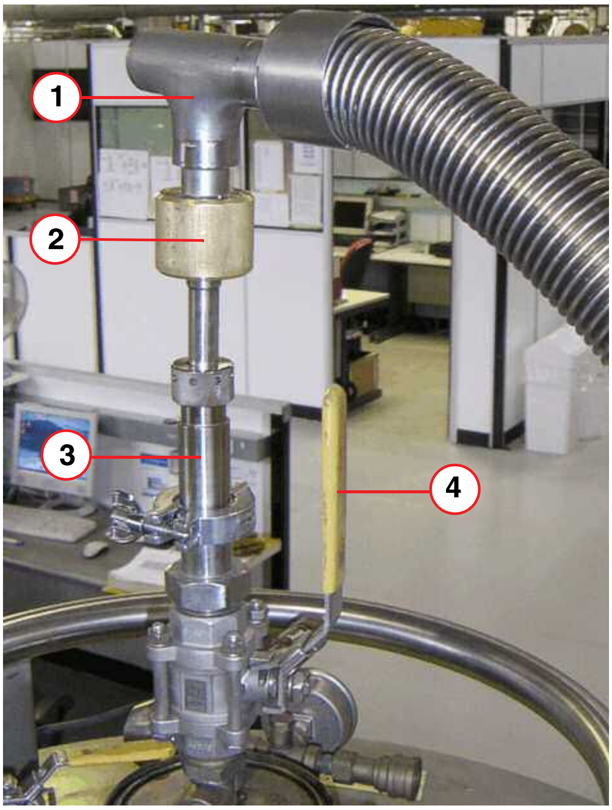
1.2.7.2 Changing supply dewars during transfer

Whistling of the syphon (a high pitched sound), or a noticeable drop in dewar pressure are indications of an empty dewar.

If the supply dewar empties before the magnet is filled, it will be necessary to replace the supply dewar with a full one.

Ensure that a full dewar is available and depressurized, ready to receive the syphon.

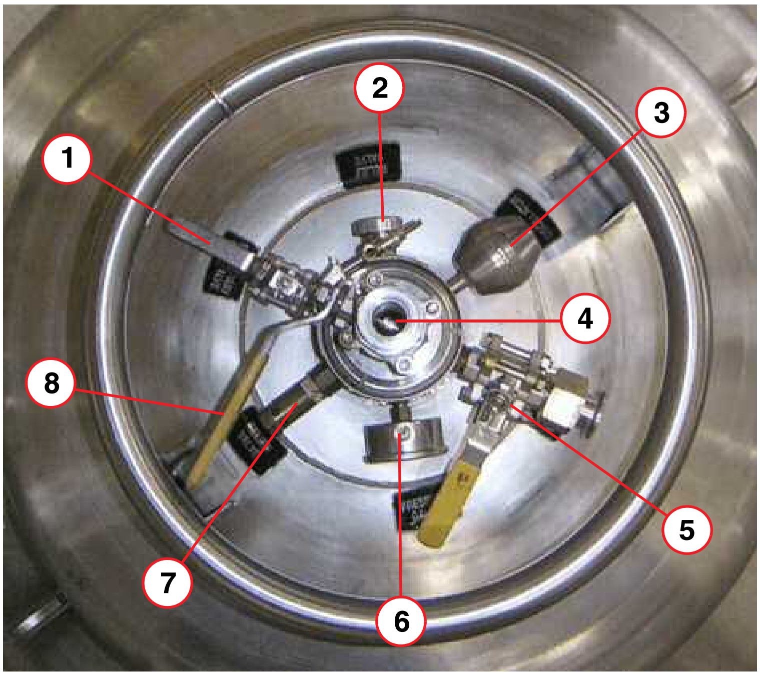
Fig. 52: Dewar syphon fitment



(1) Syphon
(2) Syphon flow valve
(3) Dewar syphon adaptor
(4) Syphon entry valve

- When the dewar empties, stop the transfer **immediately** by closing the **dewar syphon flow valve** (pos 2).
- **Close** the **primary dewar ball valve**. (→ 5/Fig. 53 Page 51)
- Turn **off** the dewar gas supply:
  - remove the **gas connection**.
  - vent the dewar down to atmospheric pressure.
  - **Ensure the area is well ventilated**.

Fig. 53: Dewar Ports (typical)



(1) Low pressure relief valve (shown open)
(2) Relief valve
(3) Anti-oscillator
(4) Syphon entry port (shown closed)
(5) Primary ball valve (shown closed)
(6) Pressure gauge
(7) Relief valve
(8) Syphon entry valve

- Using a heat gun, **warm** the **syphon connecting joint** to room temperature.
- **Disconnect** the syphons.

CAUTION

Risk of cold burns.

A "cascading plume" of cold gas will exhaust from the syphons when they are dis-connected.

» **Stand clear of the syphon joint when disconnecting.**

» **Wear protective gloves and safety glasses.**

- Remove any moisture from the end of the **service syphon** with paper towelling.
- Fit the **service syphon end cap.** (→ Fig. 46 Page 45)

CAUTION

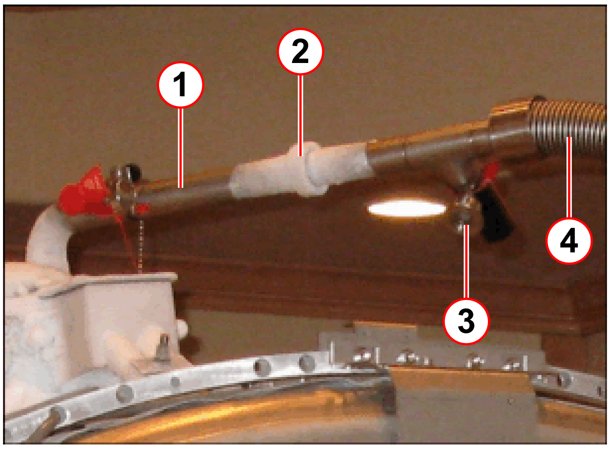
Danger of air ingress.

Internal ice blockage could cause excessive pressure.

» **Ensure the syphons are capped and plugged immediately and securely.**

- Move the full dewar close to the empty dewar.
- **Depressurize** both storage dewars down to **atmospheric pressure**.
  - » **This must be done in a well ventilated area.**
  - » **Open** the **primary ball valve.**
  - » **After** depressurizing, **close** the **valve** to avoid **air ingress and icing.**
- **Open** the **low pressure relief valve** on the **empty** dewar. (→ 1/Fig. 53 Page 51)
- **Remove** the dewar **syphon** and **adaptor** from the empty dewar and **immediately** Insert the syphon leg into the **full dewar**.
- **Close** the **syphon entry valve** (→ 4/Fig. 53 Page 51), of the **empty** dewar and **ensure the low pressure relief valve** is **opened**. (→ 1/Fig. 53 Page 51)
- **Close** the **low pressure relief valve** of the **full** dewar.
- **Purge** the gas supply line.
- **Connect** to the full dewar and **open** the dewar **primary ball valve**.
- Set the **gas pressure** to 0.27 bar (4 psi (G)).
- **Purge** the **dewar syphon of air**.
- Move the dewar next to the magnet system.

Fig. 54: Dewar and service syphon coupling



(1) Service syphon
(2) Lock nut
(3) Dewar syphon vacuum valve
(4) Dewar syphon

- Remove the service syphon end cap.
- **Reconnect** both syphons and tighten the **locking nut.** (pos 2)

- **Continue** with the transfer.

1.2.8 Stopping the helium transfer

- **As soon as** the helium level reads **95%**, or the dewar is empty, **stop the transfer** by closing the flow valve at the dewar syphon. **Do not fill above 95%.**
  - » **Whistling** (high pitched sound) **of the syphon** or noticeable **drop in dewar pressure** are indications of an **empty** dewar.
- The **dewar syphon flow valve** must be closed **immediately** when the dewar is **empty**. (→ 6/Fig. 50 Page 48)

- Set the gas **pressure regulator** to **zero**.
- **Close** the dewar **primary ball valve.** (→ 1/Fig. 43 Page 43)
- **Remove** the **gas line** from the **dewar**.
- Refer to (→ Changing supply dewars during transfer / Page 49) if the magnet is not full.

1.2.9 MSUP power

Before depressurizing the helium vessel:-

- Take a final helium level reading from the MSUP or laptop.
- Disconnect the 36 Volt power supply to the MSUP to disable the magnet pressure heater.

1.2.10 Removing the service syphon

- **Depressurize** the system helium vessel. (→ Depressurize the helium vessel / Page 37)

Fig. 55: Warm up the syphon joint with heat gun.

- **Warm** the connecting **syphon joint** to room temperature (using a **heat gun**) to remove ice.
- **Disconnect** and remove the **dewar** syphon.

CAUTION

Risk of cold burns.

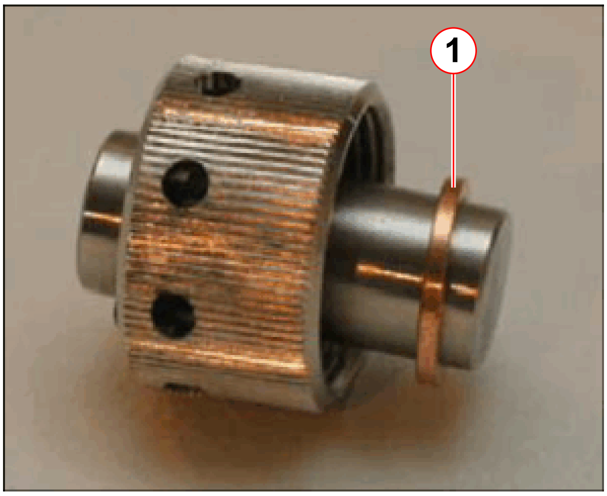
Cold gas will exhaust from the syphon port on opening.

» **Stand clear when opening.**

» **Wear protective gloves and safety glasses.**

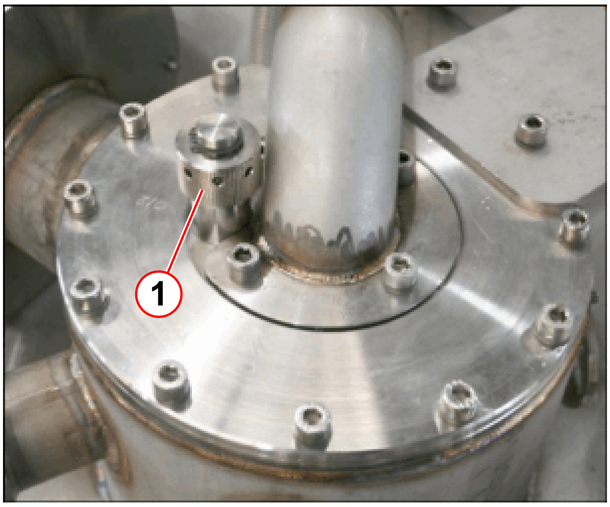
- Ensure the **syphon end cap** is **properly fitted**.

Fig. 56: Syphon blanking plug



(1) Copper seal

Fig. 57: Fit the syphon blanking plug



- **Undo** the **syphon ring nut** using the **'C'-spanner.**
- **Remove** the **service syphon** from the turret port.
- **Immediately** fit the **syphon blanking plug** using a new copper seal. (→ Fig. 56 Page 54)
- **Warm up the threads** on the plug and **carefully** tighten with the syphon 'C' spanner.
  - There might be ice on the threads - **do not use undue force**. Damaged threads will result in a leak.
- Warm the **top plate** and **syphon plug** using a heat gun. **Re-tighten** the plug.

CAUTION

Danger of air ingress.

Internal ice blockage could cause excessive pressure.

» **Warm up syphon port plug sufficiently to ensure there is no ice on the threads when tightening.**

- **RE-OPEN** the **cold head bypass valve** (standard transport configuration).
- Allow the magnet to **re-pressurize.**

  » Due to the **warmer** magnet shields at this time, during transportation, the **re-pressurization will occur naturally,** and quickly after the helium fill.

1.2.10.1 Final work steps

After the helium transfer, **check**:-

- The **magnet bypass valve** is **closed.**
- The **magnet pressure** is venting safely through the 16 or 17 psi (A) vent valve**.**
- The **turret** and **venting** system are **warm** and **dry**.

- Ensure the used **dewars** are correctly **closed**, with their low pressure relief valves **open**.
- **Move** the dewar into a well ventilated area.
  - Complete depressurizing the dewar by **opening** the **primary ball valve**.
  - When at the **gas pressure** is at **zero**, **close the primary ball valve** on the dewar.
  - **Remove** the **dewar syphon**.
  - **Immediately** close the **syphon entry valv**e on top of the dewar.
  - **Open** the low-pressure **dewar relief valve**.
  - **Warm** both syphons with the **heat gun** until dry, and store safely.
  - Remove the laptop, (if fitted).

Check the magnet pressure

- Check the **magnet pressure** is > 1.0 psi (G) **before** starting the **leak check** of the turret.

Leak check

- After the helium fill has been completed, **leak check the turret.** Refer to **Venting leak check procedure.** (→ Venting leak check / Page 66).

1.3 Removing the service syphon

- **Depressurize** the system helium vessel. (➔ Depressurize the helium vessel / Page 37)

Fig. 58: Warm up the syphon joint with heat gun.

- **Warm** the connecting **syphon joint** to room temperature (using a **heat gun**) to remove ice.
- **Disconnect** and remove the **dewar** syphon.

⚠ CAUTION

Risk of cold burns.

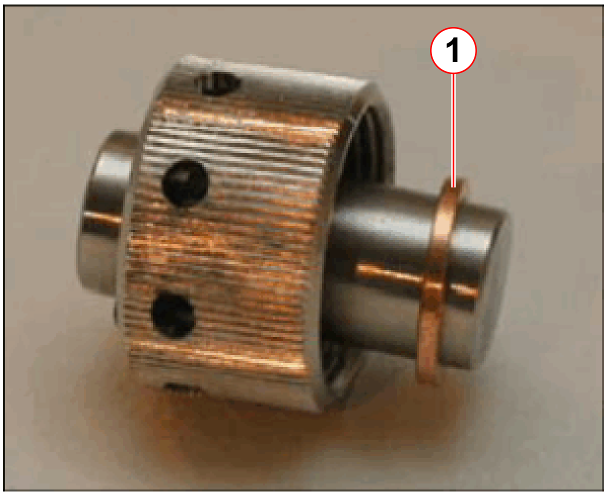
Cold gas will exhaust from the syphon port on opening.

» **Stand clear when opening.**

» **Wear protective gloves and safety glasses.**

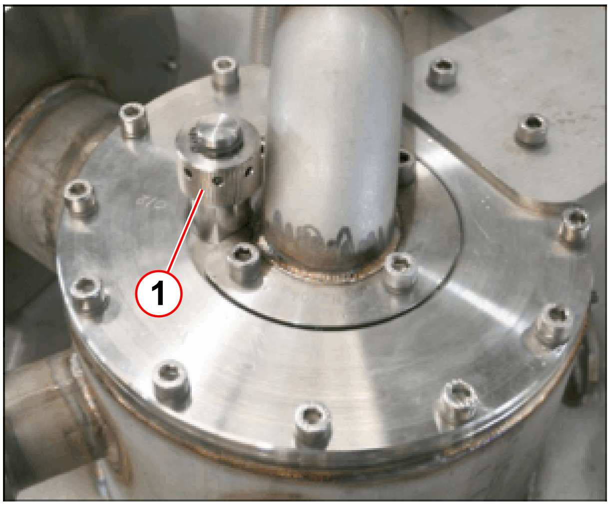
- Ensure the **syphon end cap** is **properly fitted**.

Fig. 59: Syphon blanking plug



(1) Copper seal

Fig. 60: Fit the syphon blanking plug



- **Undo** the **syphon ring nut** using the **'C'-spanner.**
- **Remove** the **service syphon** from the turret port.

- **Immediately** fit the **syphon blanking plug** using a new copper seal. (➔ Fig. 59 Page 56)
- **Warm up the threads** on the plug and **carefully** tighten with the syphon 'C' spanner.
  - There might be ice on the threads - **do not use undue force**. Damaged threads will result in a leak.
- Warm the **top plate** and **syphon plug** using a heat gun. **Re-tighten** the plug.

CAUTION

Danger of air ingress.

Internal ice blockage could cause excessive pressure.

» **Warm up syphon port plug sufficiently to ensure there is no ice on the threads when tightening.**

- **RE-OPEN** the **cold head bypass valve** (standard transport configuration).
- Allow the magnet to **re-pressurize.**
  - » Due to the **warmer** magnet shields at this time, during transportation, the **re-pressurization will occur naturally,** and quickly after the helium fill.

1.3.1 Final work steps

After the helium transfer, **check**:-

- The **magnet bypass valve** is **closed.**
- The **magnet pressure** is venting safely through the 16 or 17 psi (A) vent valve**.**
- The **turret** and **venting** system are **warm** and **dry**.
- Ensure the used **dewars** are correctly **closed**, with their low pressure relief valves **open**.
- **Move** the dewar into a well ventilated area.
  - Complete depressurizing the dewar by **opening** the **primary ball valve**.
  - When at the **gas pressure** is at **zero**, **close the primary ball valve** on the dewar.
  - **Remove** the **dewar syphon**.
  - **Immediately** close the **syphon entry valv**e on top of the dewar.
  - **Open** the low-pressure **dewar relief valve**.
  - **Warm** both syphons with the **heat gun** until dry, and store safely.
  - Remove the laptop, (if fitted).

1.3.1.1 Check the magnet pressure

- Check the **magnet pressure** is > 1.0 psi (G) **before** starting the **leak check** of the turret.

1.3.1.2 Leak check

- After the helium fill has been completed, **leak check the turret.** Refer to **Venting leak check procedure.** (→ Venting leak check / Page 66).

1.4 Bottom filling the magnet

CAUTION

Cold head failure.

The cold head seals can fail, and leak to atmoshphere.

» **Always ensure the cold head bypass valve is CLOSED during helium filling.**

Bottom filling of the magnet will be required if:-

- The liquid helium level -
  - Falls to **zero** percent,
  - **and**
  - The temperatures measured from the ceramic carbon resistors **(CCR)** and coil diodes, indicates temperatures **no warmer than** 110K. Refer to (→ Fig. 62 Page 61).

The magnet must be filled through the **magnet syphon cone** to allow filling from the bottom of the helium vessel.

> Inform Magnet HSC immediately if the helium level is zero. (→ Magnet HSC, mailto:magnet.wscshsc.healthcare@siemens.com)

Possible causes for this include:-

- Faulty helium level probes.
- **Long** transportation and storage **times**.

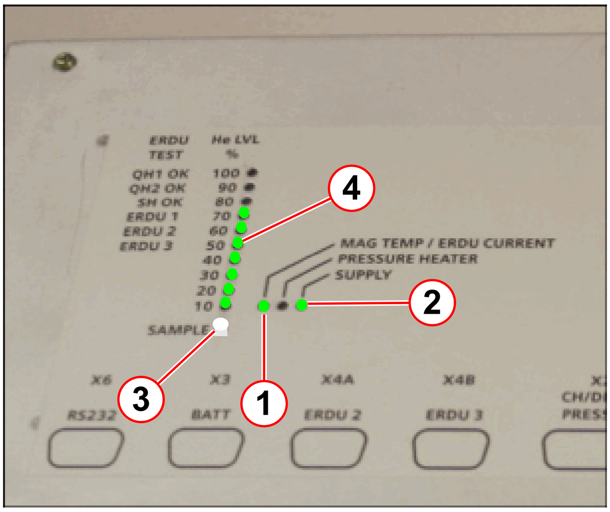
1.4.1 Checking magnet temperatures

To **recover** a magnet with a **bottom helium fill**, the following must be checked:-

- **Helium level** values.
- **Magnet temperature CCR** values.

> Magnet coil CCR temperatures MUST BE CONFIRMED before continuing with the helium fill.

Fig. 61: Check the magnet temperature



(1) MAG TEMP/ERDU CURRENT
(2) SUPPLY
(3) SAMPLE BUTTON
(4) He LVL %

- Check the **MAG TEMP LED** (pos 1) (➔ Tab. 13 Page 60)

Tab. 13 MSUP 2306 - Magnet temperature indicators

Temperature	LED status	Action	
Room to nitrogen (77K)	RED	Check the CCR temperatures	Contact **magnet HSC IMMEDIATELY**
Nitrogen to helium	YELLOW	Refill **immediately**	Bottom fill
Helium temperature normal (4.2K)	GREEN	Refill as required	Top fill

Fig. 62: OR98 Magnet Acceptance Document (example)

SIEMENS Magnet Acceptance Documentation
Installation Data

MR Magnet Technology Quality Regulations

Cryostat Number 21236477 **SMT Reference** 21245226 05.10

System Description

1.495T Actively Shielded Superconducting Twin Turret Magnet System OR98

Measured Magnet Centre (Offset)	+1 (P)	mm
Gradient Coil Position	39	mm
Overall System Length	1409	mm

This system requires 12 Z2 rings and 2.38 Kg of iron to shim. This is a theoretical calculation that may change due to site conditions.

Operating Parameters

Description	Measured value	
Operating Current @ 1.495T	510.83	AMPS
Field/Current Ratio	12.46	kHz/0.1 Amp

Magnet Type: OR98--??

Carbon Resistor Values

POSITION	Room Temperature		Helium Temperature
CCR4 TDC Outer (D-E	926		2951
CCR3 Top Inner (C-E	1060		2812
CCR2 Middle Inner (B-E)	1035		2977
CCR1 Bottom Inner (A-E	1093		3138
TDC Calibration Values:	882.4	1181.8	3075.3

Helium Level Probe Resistance (When using a 250 mA supply)

	Probe 1	Probe 2	
0% Resistance	107.0	107.0	Ohms
0 % Calibration	448	448	
100 % Resistance	17.9	17.9	Ohms
100 % Calibration	75	75	
Diode Table No	-2		
Pressure Heater Res	22.1		Ohms

Final Temperature Sensor Recon Values

50Kbore	50K Link	Coldhead		Recon Margin
56	55	41	K	782 Milliwatts

Checked by K Suraina

Signature

Date 28/05/2010

1.4.2 Bottom fill procedure

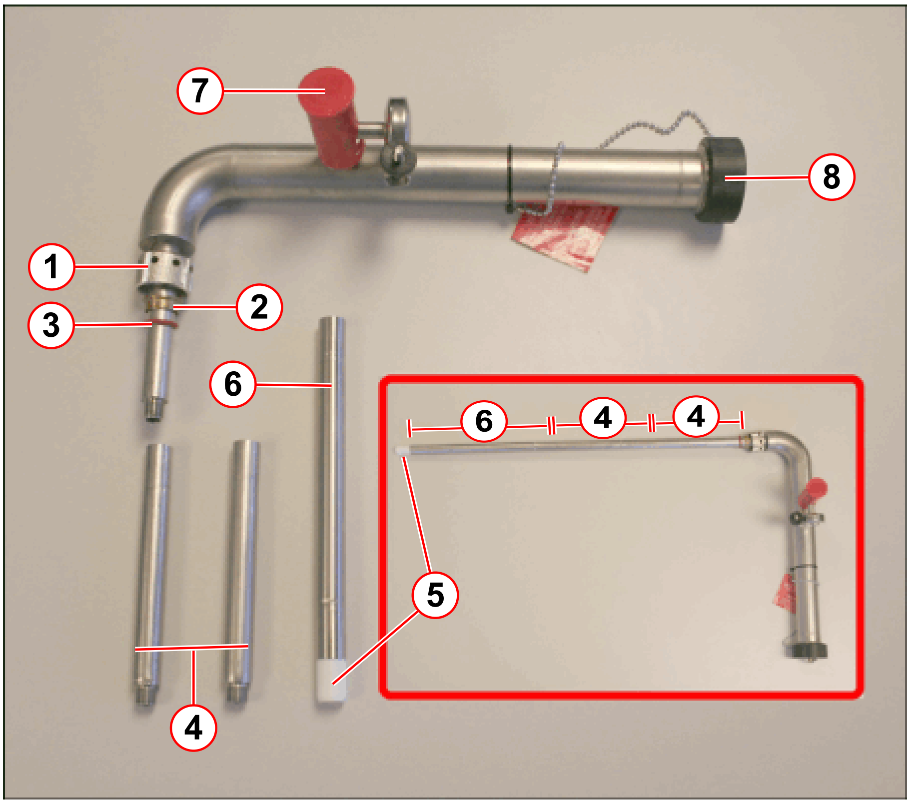
Follow the previous steps for **Safety**, **Site** and **System preparation.**

- Prepare the **dewar** and **dewar syphon.** (→ Prepare the helium dewar and equipment / Page 39)

1.4.3 Assemble the service syphon (bottom filling)

1.4.3.1 Assemble the service syphon

Fig. 63: Magnet service syphon assembly - bottom filling

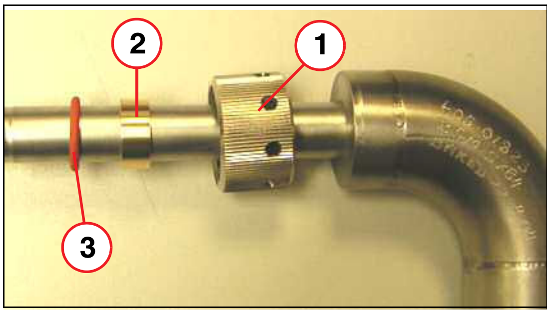


(1) Ring nut
(2) Brass spacer
(3) O-ring seal
(4) 140 mm syphon legs (2 off)
(5) 22 mm PTFE seal (bottom fill)
(6) 200 mm syphon leg (1 off)
(7) Vacuum valve
(8) Syphon end cap

Tab. 14 OR98 / OR99 / OR103 BOTTOM filling syphon extension legs

Magnet type	130* mm leg	190* mm leg	22 mm PTFE seal
OR98, OR99, OR103	2	1	1
Note: *Effective extension leg length			

Fig. 64: Assembling the Ring Nut Seal



(1) Ring nut
(2) Brass spacer
(3) Silicon O-ring

- Fit the ring nut, brass spacer and the O-ring to the **service syphon leg.**

- Connect the **extension legs** and the 22 mm **PTFE seal:**
  - Screw the **2** x **140 mm legs** together, and connect to the **syphon.**[1]
  - Fit the 1 x **200 mm leg.** [2]
  - Fit the **22 mm PTFE seal** to the 200 mm leg.

> The minimum length required to reach the syphon cone is:
> OR98 - 554 mm (+/- 5 mm)
> OR99/OR103 - 540 mm (+/- 5 mm)

Fit the magnet service syphon

- **Disconnect** the **36 Volt power** to the 2306 **MSUP** to disable the magnet pressure heater during depressurization.
  - Connect the 36 Volt power when helium fill has started to allow helium readings to be monitored.
- **Depressurize** the helium vessel.(→ Depressurize the helium vessel / Page 37)
- **CLOSE** the magnet **bypass valve**.
- **Warm** and dry the **syphon blanking plug** using a heat gun to ease removal.
- **Remove** the syphon **end cap.** (→ Fig. 46 Page 45)

1. When connected, the effective length of the legs is 130 mm.
2. When connected, the effective length of the legs is 190 mm

- **Remove** the syphon **blanking plug.**

CAUTION

Risk of cold burns.

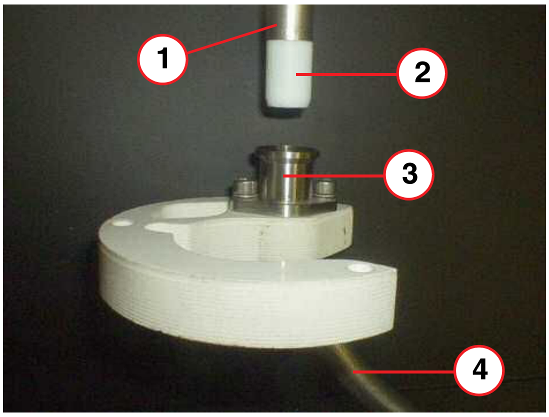
Cold gas will exhaust from the syphon port on opening.

» **Stand clear when opening.**

» **Wear protective gloves and safety glasses.**

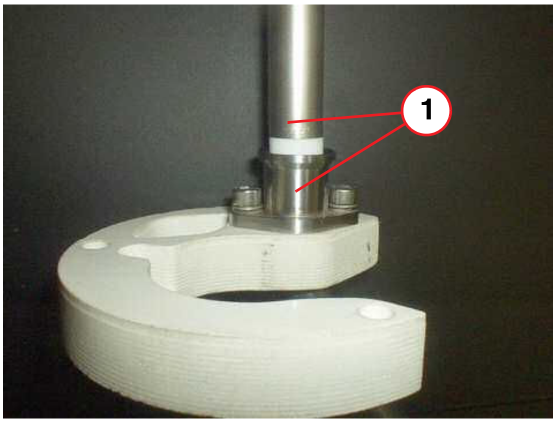
- **Dry** the syphon **port**.
- **Slowly** lower the **service syphon** fully into the neck.
- The **syphon leg** must locate fully into the **syphon cone** - make sure it is **fully engaged**. After filling a maximum of 500 ltrs of helium, the magnet should be cold enough to start collecting liquid helium.

Fig. 65: Bottom fill syphon cone (pictorial view)



(1) Syphon leg
(2) PTFE Seal (Bottom Fill)
(3) Syphon cone
(4) Syphon tube

Fig. 66: Bottom fill - syphon leg fully engaged (pictorial view)



(1) Syphon leg - fully engaged

Helium efficiency will be reduced if the syphon leg is not fully engaged in the syphon cone.

- Tighten the **syphon port ring nut** (→ 1/Fig. 64 Page 63) by hand, then apply a half turn, (using the appropriate spanner), to form an air-tight seal.
- Re-fit the **syphon end cap.**

1.4.4 Liquid helium transfer

To prepare and fill the magnet with helium, follow the procedure **Liquid helium transfer** (→ Liquid helium transfer - top fill / Page 47)

1.4.5 Monitor the magnet CCRs

- Monitor the **magnet CCRs** every **15 minutes**, to measure the **temperature drop**.
  - » Refer to the **acceptance documentation** (Installation Protocol) for required values.
  - » If the magnet CCRs **do not** get cooler, **contact Magnet HSC.** (→ Magnet HSC, mailto:magnet.wscshsc.healthcare@siemens.com)
- When the required value is reached, **monitor** the **helium level** at the **MSUP**.

> After filling a maximum of 500l helium, the magnet should be cold enough to start collecting liquid helium.

1.4.6 Final work steps

See (→ Final work steps / Page 65)

1.5 Venting leak check

1.5.1 General

For all recondensing systems, it is essential that the magnet system is leak tight to maintain a zero-loss cryogenic performance.

Individual leaks must be 1 x 10^{-4} mbar l/s^{-1} or less.

Leak checking must be performed with an electronic leak detector capable of detecting leaks, at least as small as **1 x 10^{-4} mbar l/s.**

The magnet must be checked after any **service operations** have been performed, such as:-

- After **helium filling**.
- After any **replacement** or **disturbance** of a component on the **high pressure** side of the magnet.

CAUTION

Risk of icing

Fine leaks may not show any change in pressure. Over time, they will contribute to air ingress and icing.

» **Leak check with a suitable leak check device.**

1.5.2 Leak check equipment

- **Accepted** leak detection equipment.

Fig. 67: Edwards Leak detector

- Use the Edwards hand-held leak detector (part no **10614850**).

  This is the **recommended equipment** for leaks **greater than 1 x 10^{-4} mbar l/s.**

NOTE: The hand-held device is calibrated in **ml/sec**:-

1 x 10^{-4} mbar l/s = 1 x 10^{-4} ml/sec at normal pressure and temperature.

The mass spectrometer is used for the detection of ultra fine leaks less than 1 x 10^{-5} mbar l/s. This equipment must only be used when trouble-shooting is required.

- Check that the equipment is detecting helium.
  - Check the equipment using helium from a known gas supply (i.e. regulated gas bottle).
  - Refer to the equipment manufacturer's operating instructions.
  - The calibration certificate must be valid.

1.5.3 Pre-requisites for leak checking

- The helium vessel must have a **positive** pressure > 0.5 psi (G) prior to leak checking.

Tab. 15 Magnet pressure OR98/OR99/OR103

	OR98	OR99/OR103
Magnet pressure	16 psi (A) 1.3 psi (G)	17 psi (A) 2.1 psi (G)
Typical range at **sea level** pressure gauge reading (atmospheric variation)	0.7 - 2.3 psi (G)	1.7 - 3.3 psi (G)
Magnets installed at 2000 m (example)	4.5 psi (G)	5.2 psi (G)

- Leak check all **seals** that have been subjected to **prolonged low temperature** and the subsequent build up of frost. For example, those which have been cooled to low temperature by helium transfers, helium vessel de-pressurization, or magnet ramping.
  - Any **joints** which are broken and re-made during servicing must be leak checked.
  - The **turret, 16 psi (A) or 17 psi (A) vent valve assembly** and quench valve must be **warmed up to room temperature** after any service procedures have been performed, prior to leak check.

1.5.4 Procedure

> i The leak detector must be set for maximum sensitivity.

- Switch on the leak detector and allow it to warm up (refer to manufacturer's instructions).
- Check the equipment is working correctly (detecting helium). This should be done both before and after use.

> i Ensure that the atmospheric environment is not saturated with helium gas (i.e. do not leak check soon after a helium fill, or depressurization).

The background must be at least one order of magnitude lower than the smallest leak expected, (i.e. leaks of 10^{-4} mbar l/s, require background of order 10^{-5} mbar l/s).

This level will be undetectable by the hand-held device.

- Note response time of the detector.

During leak checking, move the instrument slowly over the leak point to give the detector time to respond.

Start at the **lowest** point:

- Hold the detector at the suspected leak point until the reading has stabilized.
- Move the detector away and check that the indication disappears.
- Repeat the process to check that a leak is still indicated.

The following points must **always** be observed:-

- Joints on the **'HIGH' pressure** side of the venting system will generally be fitted with **metal seals.** Before dismantling a joint, **ensure a replacement seal is available.**
- **DO NOT** use rubber seals as a substitute for metal seals.
- **DO NOT** re-use metal seals, always replace with a new seal.
- **DO NOT** use vacuum grease.

Do not use any form of vacuum grease on O-rings and seals.

It is recommended that these be wiped clean and checked for damage prior to fitting. Vacuum grease dries over time and can cause a seal to leak.

- After repair, leak check the entire venting system, not just the joints which have been repaired.

1.5.5 External leak checks

1.5.5.1 External leaks

The venting and components on the **high pressure** side of the magnet system require leak checking to ensure that the re-condensing performance is achieved. This includes:

- Cold head (→ Cold head checks / Page 69).
- Turret top plate, including helium syphon port and venting. (→ High pressure turret and venting joints / Page 71)
- Vent valve assembly (→ Vent valve assembly / Page 72)
  - **16** psi (A) = **OR98**
  - **17** psi (A) = **OR99** and **OR103**
- Pressure gauge assembly. (→ Pressure gauge / Page 72)

- Connecting pipe couplings.

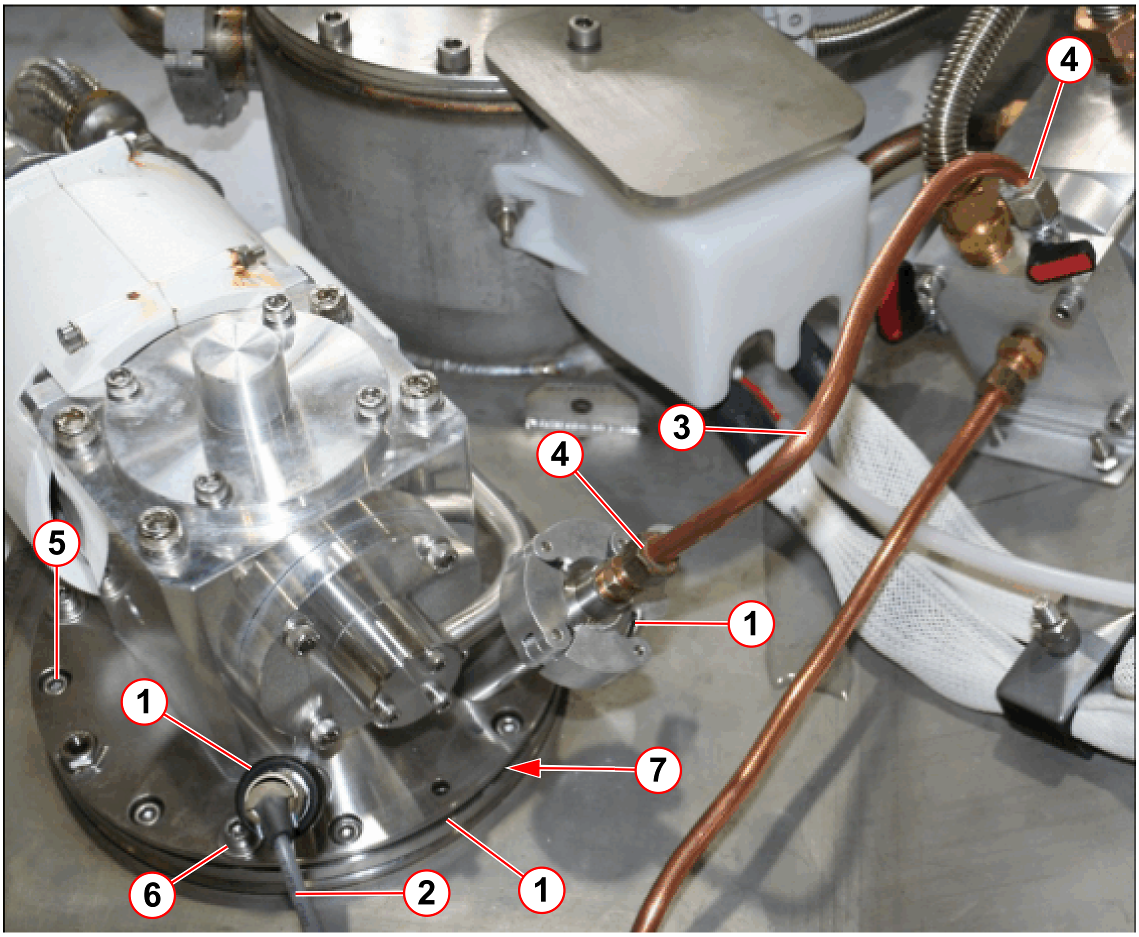
> If no external leak is found but the system pressure is low, then it is possible that one of the internal valves is leaking. This can be caused by a cracked or broken burst disk, or a broken O-ring.

If leaks are found at **vent line connections**, first tighten the clamp, or securing screws and re-check. **DO NOT OVERTIGHTEN**.

If this does not cure the leak, **escalate to Magnet HSC**. High load seals may need to be replaced. (➔ Magnet HSC, mailto:magnet.wscshsc.healthcare@siemens.com)

1.5.5.2 Cold head checks

Fig. 68: Leak checking the cold head seals

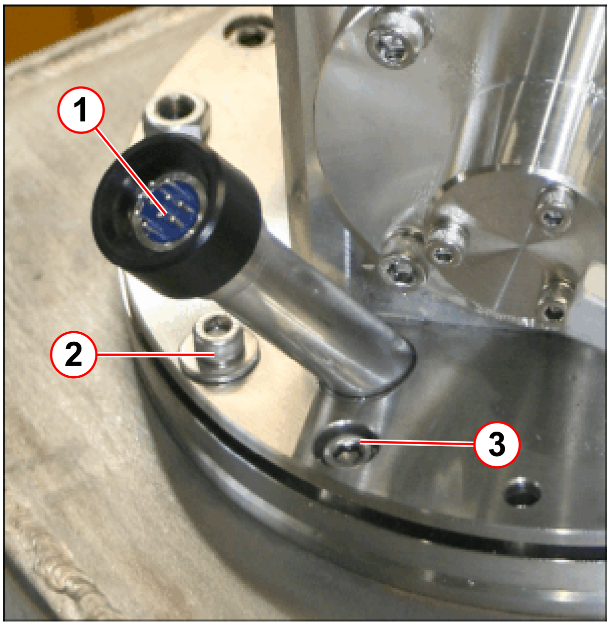


(1) Cold head seals leak check points
(2) Temperature sensor cable
(3) Cold head vent line
(4) Cold head bypass line connection
(5) 8 off M6 x 16 mm O-ring clamp ring screws, torque to = 8Nm
(6) 4 off M6 x 25 mm cold head flange screws, torque to = 4Nm
(7) O-ring seal

- The **cold head** is sealed in position by an **O-ring**. (➔ 7/Fig. 68 Page 69)

- Check the **cold head vent line seals** at both ends. (➔ 4/Fig. 68 Page 69)

Fig. 69: Cold head 10 pin seal



(1) Leak check point
(2) 4 off M6 x 25 mm cold head flange screws, torque to 4Nm
(3) 8 off M6 x 16 mm skt hd screws O-ring clamp plate, torque to 8Nm

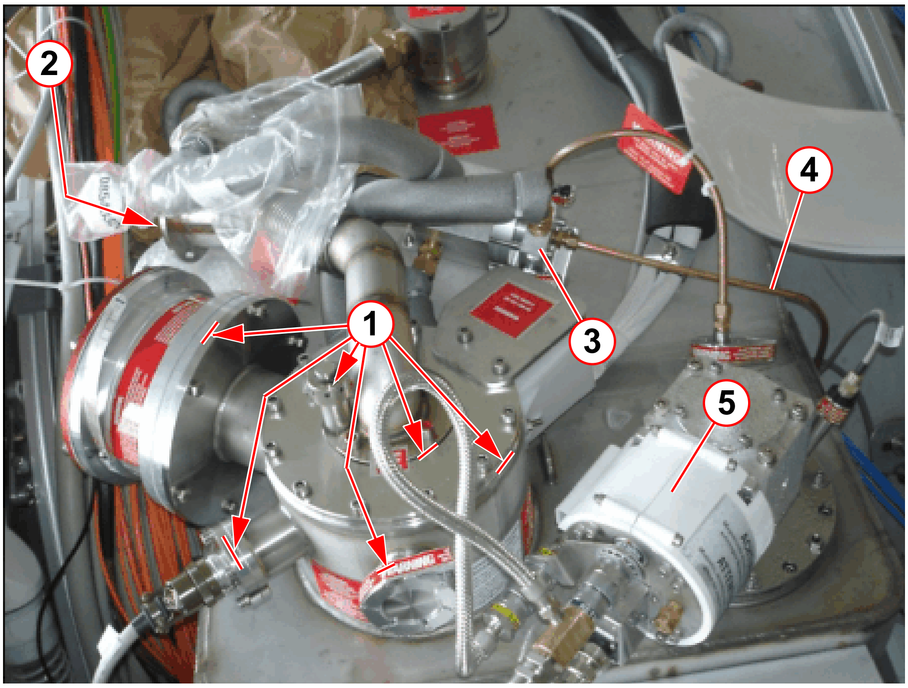
- Remove the **temperature sensor cable**.
- Leak check the **10 pin seal.**

The cold head is fitted with **temperature sensors**:

- The **feed-through** for the wires is sealed with an **O-ring**.
- Check the **feed-through** and **cold head seals.**
- **Check** the **torque** of the cold head **screws**.

High pressure turret and venting joints

Fig. 70: Turret high pressure joints



(1) High pressure joints
(2) Aux vent burst disk
(3) Vent valve assembly
(4) Pressure gauge pipe
(5) Cold head

High pressure leak checks

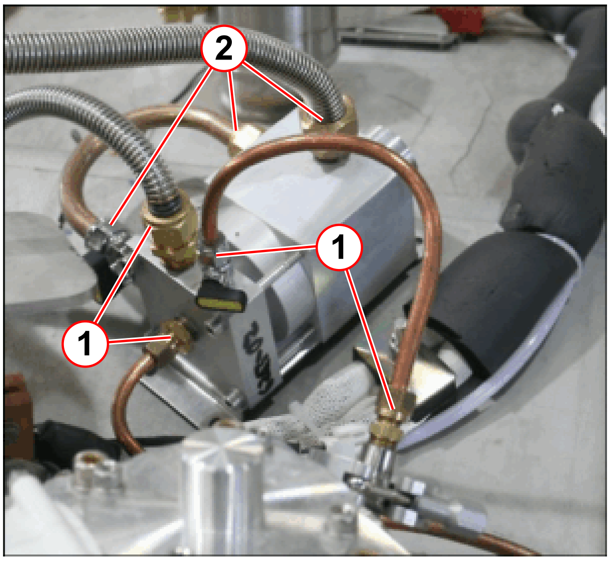
- **Check** that all **clamps** on the **high-pressure side** of the venting are tight. **Do not** try to **over-tighten** them - it could cause a leak, or make an existing leak worse.

> Replace a high load seal only with another identical high load seal. High load seals must only be used ONCE for a guaranteed leak tight seal.

- Check the **syphon port** and other seals on the **turret top plate**.
  - Use a **heat gun** to warm up and dry the **turret** and **syphon port**.
  - leak check the **syphon port**.
- **Connections** to the **vent valve assembly** - see also (→ Vent valve assembly / Page 72)
- **Cold head** line - see also (→ Cold head checks / Page 69)
- **Pressure gauge** line - see also (→ Pressure gauge / Page 72)

Vent valve assembly

Fig. 71: Vent valve assembly leak check points

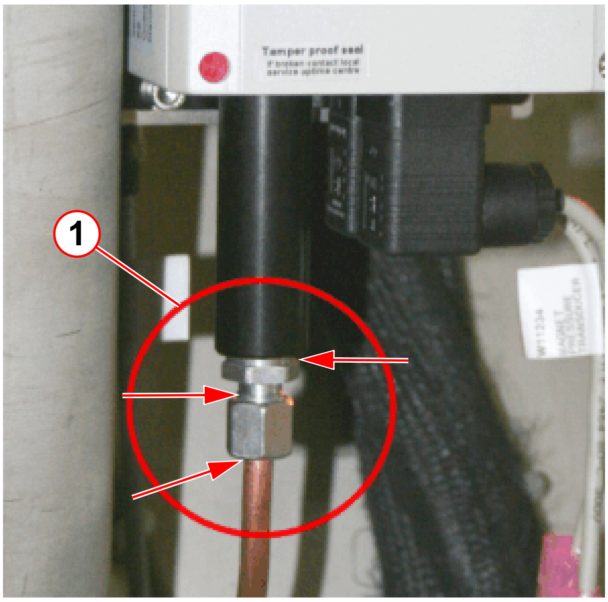


(1) High-pressure joints
(2) Low-pressure joints

- Leak check the vent valve **high pressure** assembly.
- For leaks found at pipe couplings, first try **tightening** the clamp or securing screws. **DO NOT OVER-TIGHTEN!**
- If tightening does not solve the leak, the **complete** vent valve assembly must be replaced. (Refer to **Replacement of Parts**).

Pressure gauge

Fig. 72: Leak check the pressure line



(1) Leak check points

Leak check the **high pressure points**.

- For **leaks** found at pipe couplings, first try tightening the clamp or securing screws. **BEWARE OF OVER-TIGHTENING!**
- If tightening does not solve the leak, the **metal seal** must be **replaced**. (Refer to **Replacement of Parts**)
- Contact your local **Uptime Service Centre** if further leaks are suspected.

1.5.6 Internal leaks checks

- Perform a visual check of the auxiliary **1.38 bar** metal **bursting disk**.
  (→ 1/Fig. 73 Page 73)

CAUTION

Internal leak checks.

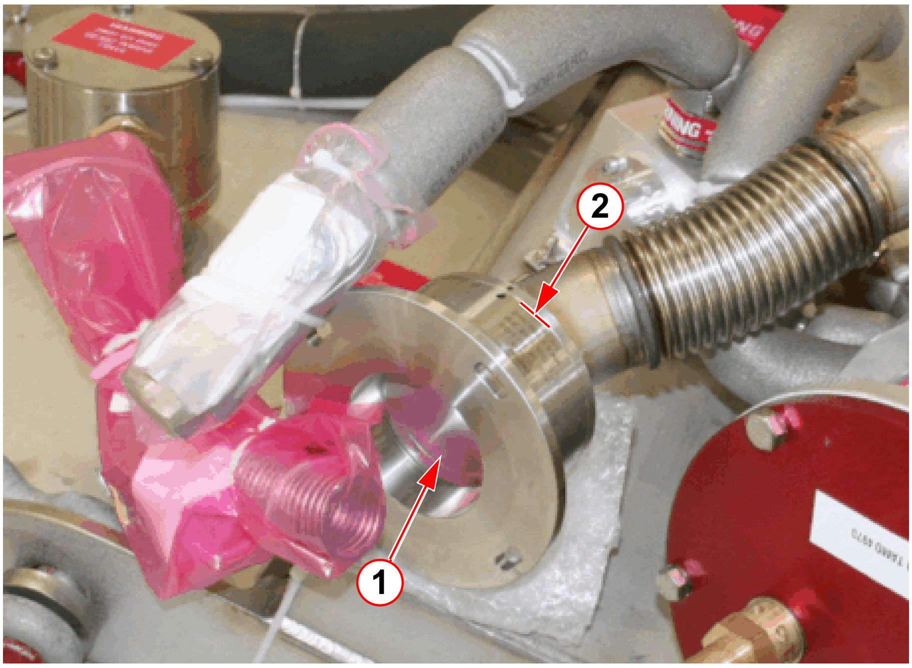
Risk of air ingress causing internal icing of turret.

» **Replace burst disk or valve immediately if found to be broken or leaking.**

1.5.6.1 Auxiliary vent 1.38 bar metal bursting disk

The membrane is very fragile - DO NOT TOUCH THE SURFACE.

Fig. 73: Leak check the auxiliary burst disk



(1) Visual check of the burst disk
(2) Aux burst disk high pressure joint

- **Visually inspect** the auxiliary **burst disk membrane** to ensure that it is not damaged.
- **Replace** the auxiliary metal burst disk if a **leak is detected.**

A broken auxiliary burst disk MUST be escalated to Magnet HSC. (→ Magnet HSC, mailto:magnet.wscshsc.healthcare@siemens.com)

2.1 Checking the Ceramic Carbon Resistors (CCRs)

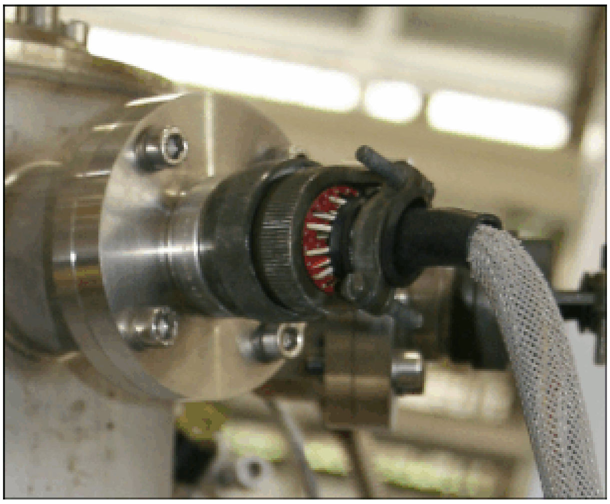
This procedure may only be performed by a Siemens CSE.

Tools required

- Multimeter and pin leads
- Laptop or PC with terminal emulation software
- RS232 cable

- Check the **Ceramic Carbon Resistors (CCRs)** from the **MSUP** display screen or by using a **multimeter**. See also (➔ Using the Laptop to check magnet parameters / Page 77).
- The measurement is taken in **Ohms**, across the **pins** on the **41pin turret connector**.

Fig. 74: Turret - 41pin connector



- **Disconnect** the **41pin connector**.
- **Measure** across pins **a-e**, **b-e**, **c-e**, **d-e**.
- **Compare** the **CCR temperature values** with the **Magnet Acceptance Document**.[1](➔ Fig. 76 Page 76)
- **Re-connect** the **41pin connector**.

---

1. OR98 shown

Tab. 16 Expected CCR resistor values

41 way plug, measured in Ohms	Expected values room temperature	Expected values 77 K temperature	Expected values 4.2 K temperature
All ceramic carbon resistors (CCRs)	880 - 1050	1150 - 1350	2800 - 3380

Fig. 75: 41pin turret connector - ceramic carbon resistor pins (room temperature values)

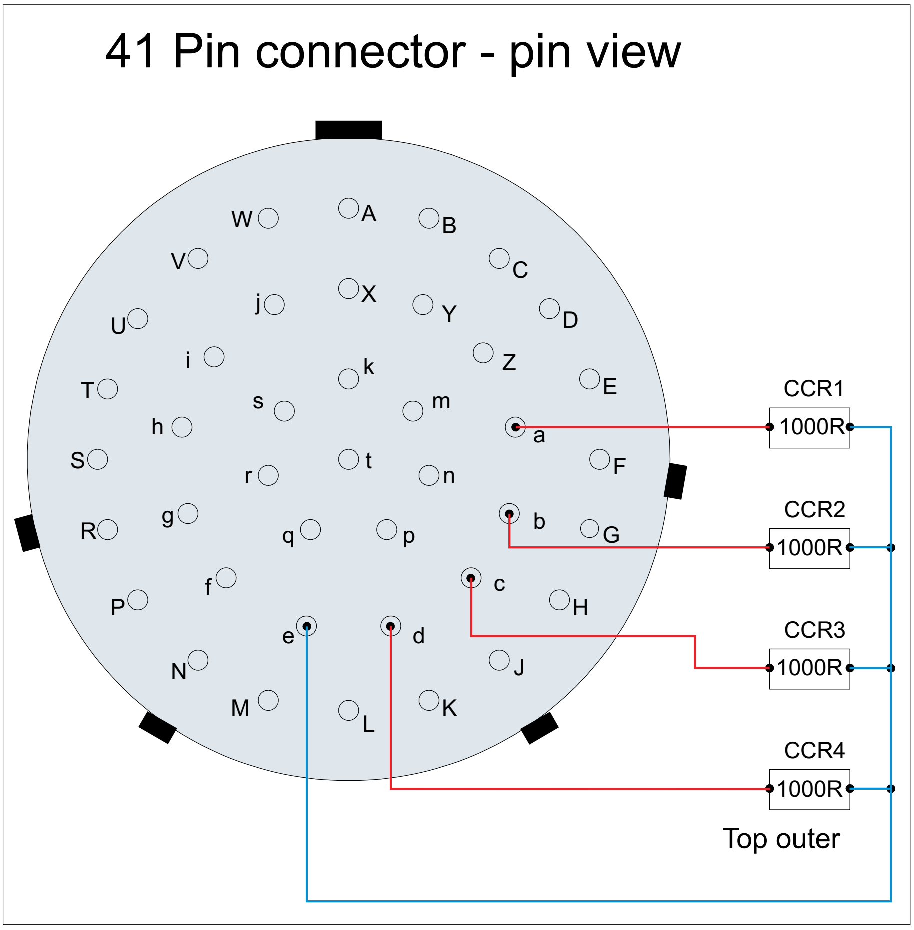


Fig. 76: OR98 Magnet Acceptance Document (example)

SIEMENS

Magnet Acceptance Documentation
Installation Data

MR Magnet Technology Quality Regulations

Cryostat Number 21236477 **SMT Reference** 21245226 05.10

System Description

1.495T Actively Shielded Superconducting Twin Turret Magnet System OR98

Measured Magnet Centre (Offset)	+1 (P)	mm
Gradient Coil Position	39	mm
Overall System Length	1409	mm

This system requires 12 Z2 rings and 2.38 Kg of iron to shim. This is a theoretical calculation that may change due to site conditions.

Operating Parameters

Description	Measured value	
Operating Current @ 1.495T	510.83	AMPS
Field/Current Ratio	12.46	kHz/0.1 Amp

Magnet Type: OR98--??

Carbon Resistor Values

POSITION	Room Temperature		Helium Temperature
CCR4 TDC Outer (D-E	926		2951
CCR3 Top Inner (C-E	1060		2812
CCR2 Middle Inner (B-E)	1035		2977
CCR1 Bottom Inner (A-E	1093		3138
TDC Calibration Values:	882.4	1181.8	3075.3

Helium Level Probe Resistance (When using a 250 mA supply)

	Probe 1	Probe 2	
0% Resistance	107.0	107.0	Ohms
0 % Calibration	448	448	
100 % Resistance	17.9	17.9	Ohms
100 % Calibration	75	75	
Diode Table No	-2		
Pressure Heater Res	22.1		Ohms

Final Temperature Sensor Recon Values

50Kbore	50K Link	Coldhead		Recon Margin
56	55	41	K	782 Milliwatts

Checked by K Suraina

Signature

Date 28/05/2010

2.2 Using the Laptop to check magnet parameters

A Siemens password is required to access the full MSUP command screens.
Only Siemens CSE's have access to the password.

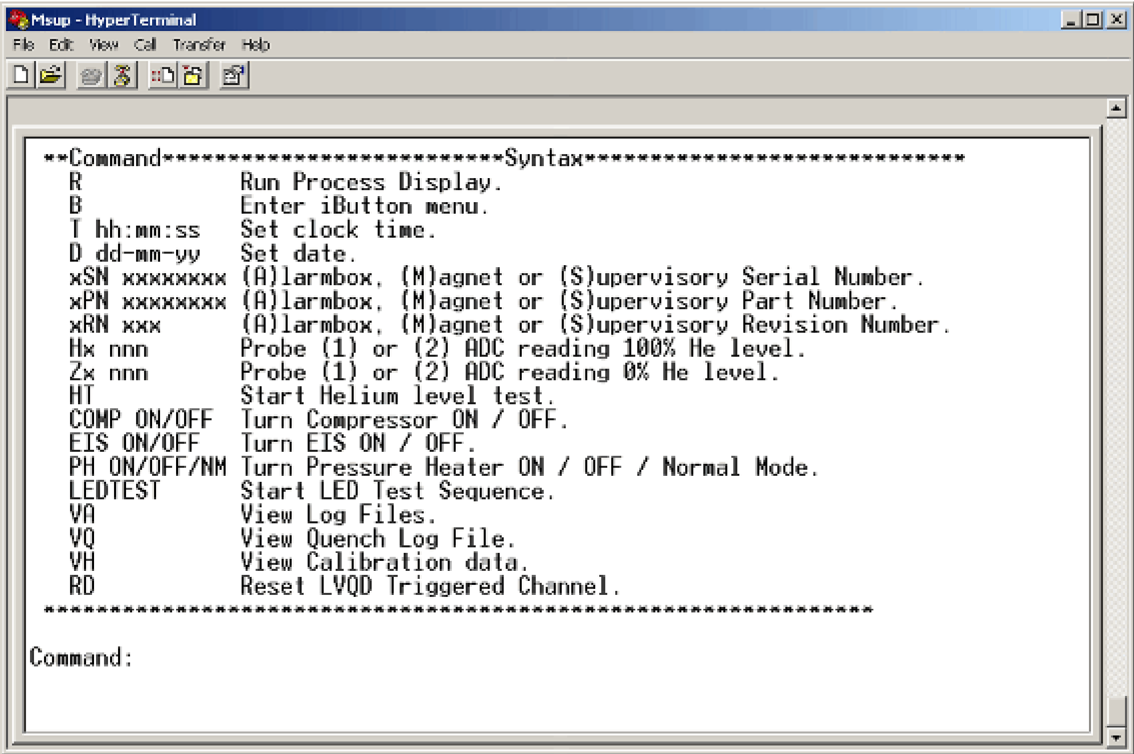
To check magnet parameters:

- Turn on the **laptop** and open the **HyperTerminal application.**
  - **Connect** the laptop to connector **X6** on the MSUP via **RS232** cable.
  - **Press <ESC>** to display the initial **MSUP Command input screen**.

If the screen freezes, press ESC or REFRESH.

  - Enter the **Siemens CSE password** to allow full access to the **MSUP Command input screen.**

Fig. 77: Command input screen



Checking the CCR temperatures from the MSUP

- From the MSUP Process display screen check **CCR sensor 4.**
- Compare this value with the **Magnet Acceptance Document**.

2.3 De-icing the syphon port

Only trained magnet CSEs are allowed to perform de-ice procedures.

If the syphon port is iced, escalate to Magnet HSC IMMEDIATELY. (→ Magnet HSC, mailto:magnet.wscshsc.healthcare@siemens.com)

2.3.1 General Information

The service syphon is designed to be fully portable and compatible with all Siemens **4K** magnet products. By using a combination of the service syphon extension legs supplied, a helium top fill, bottom fill, or de-ice of the syphon port can be performed.

A de-ice may be required if:

- A **resistance** is felt when initially inserting the syphon leg.
  - » this suggests the syphon cone is iced-up.
- Poor **transfer** efficiency.
- **No movement** on the helium level **meter** over a period of time.

The **de-ice tool** is designed to blow helium gas (at room temperature), in defined directions within the syphon port, **dispersing any ice which has formed**.

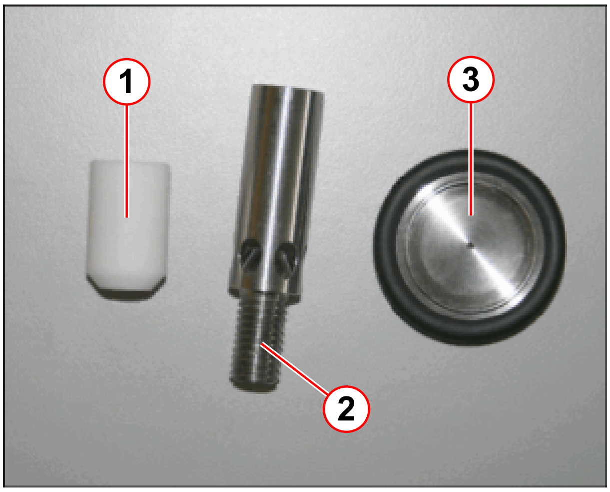
2.3.2 Requirements

Tooling

Tab. 17 Service syphon kit and upgrades

Part number	Revision	Comment
08396363	00	Service syphon kit
10126691	00	Upgrade kit (200 mm extension leg added).
10129438	00	Upgrade kit (de-ice tooling)
10126651	01	Upgrade Service syphon kit (includes upgrade kits)

Fig. 78: Upgrade kit 10129438 - De-ice tooling



(1) PTFE tip
(2) De-ice helium fill diffuser
(3) NW25 flow restrictor

2.3.3 Procedure

The **de-ice diffuser tip** is designed to direct the warm gas downwards past the syphon cone **to clear ice from the lower regions of the turret.**

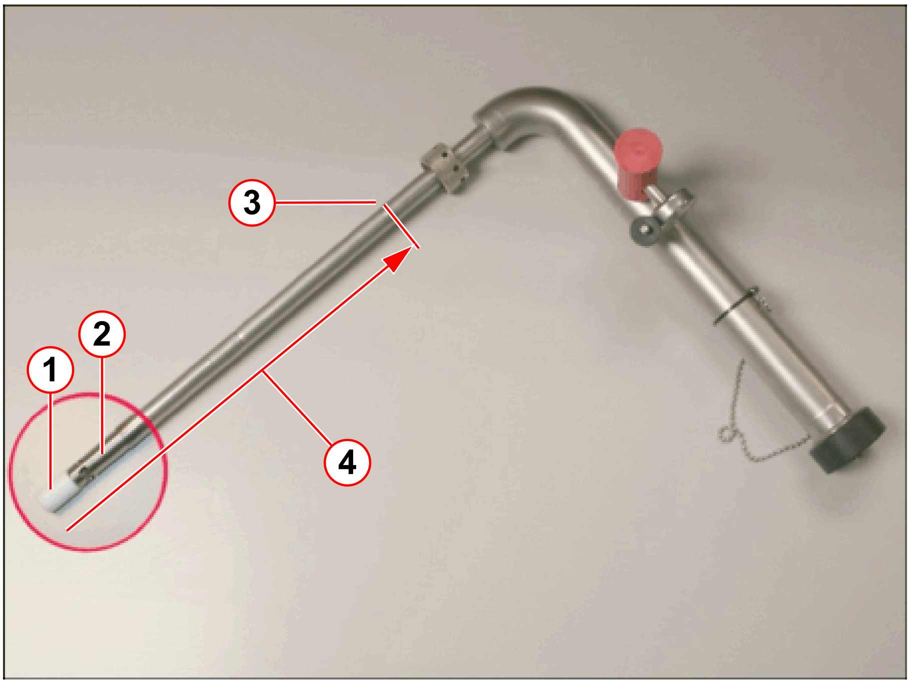
When the **syphon port is clear of ice** the diffuser tip will be able to fit into the **syphon cone**.

- **Open** the magnet **bypass valve** to **depressurize the** helium vessel.

2.3.3.1 Assemble the syphon

- Fit the required extension legs. (➔ Tab. 18 Page 80)
- Fit the **de-ice diffuser tip** to the syphon leg.
  - Check that the **PTFE extension tip** is screwed tightly onto the de-ice diffuser. (➔ 1/Fig. 79 Page 80)

Fig. 79: Syphon assembly for de-icing the syphon port



(1) PTFE tip
(2) 6 hole de-ice dffuser
(3) Syphon depth guideline
(4) Depth to cone measurement**

Tab. 18 Extension legs and de-ice diffuser requirements

Magnet Type	140 mm Leg	200 mm Leg	60 mm De-ice diffuser and PTFE tip	Depth to cone**
OR98	2	1	1	554 mm
OR99	2	1	1	554 mm
OR103	2	1	1	554 mm

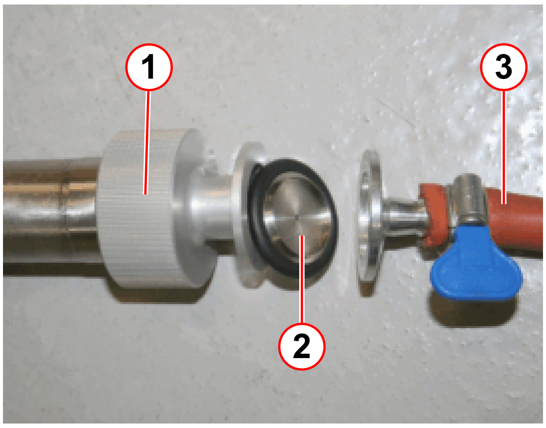
Refer to table above to determine the **depth to cone measurement** for required magnet type.

- Select the **depth to cone measurement.** (→ Tab. 18 Page 80)

> i The depth measurement from the top of the ring nut to the syphon cone on the magnet is a guide only.
>
> Add an additional 10 mm to ensure the extension leg will reach the cone.

- Mark the syphon leg with a marker pen. (→ 3/Fig. 79 Page 80)
  - » **Example:** OR98: Mark a line at **554 mm** (554 mm + 10 mm).
- **Remove** the syphon **end cap**.

Fig. 80: NW25 adaptor and flow restrictor

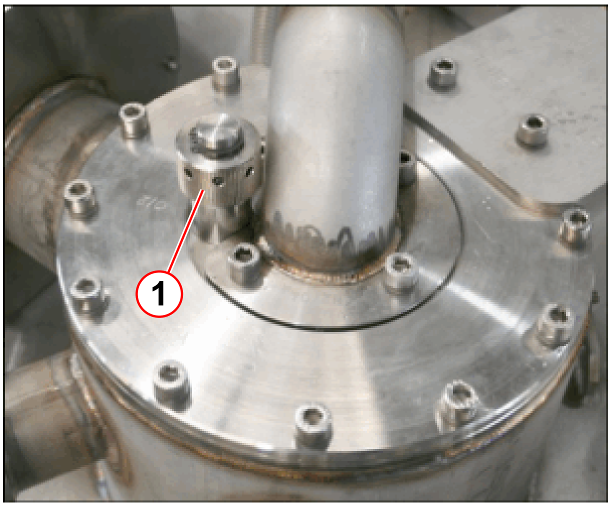


(1) NW25 fill adaptor
(2) NW25 flow restrictor
(3) Helium supply line

- Fit the **NW25 adaptor** to the coupling end of the syphon.
- Fit the **NW25 flow restrictor**, between the **adaptor** and helium **supply line.**
  - » The NW25 restrictor will allow a **calibrated flow of gas** to pass through it for de-icing.

- **Connect** the high pressure **regulator** to the helium gas bottle, 4.6 grade (99.996% (minimum)) helium gas.
- **DO NOT** turn on the helium gas supply at this point.
- Confirm the magnet is **depressurized completely.**
- **Close** the magnet **bypass valve.**

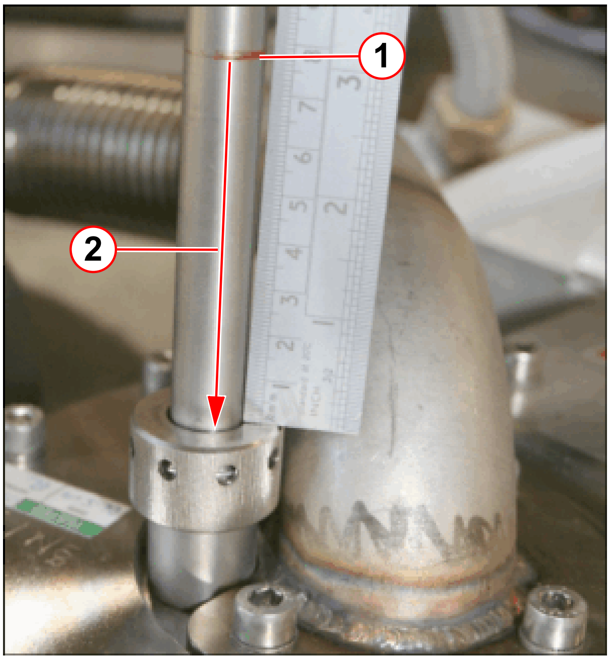
Fig. 81: Syphon entry port



(1) Syphon blanking plug

- Warm and dry the **syphon blanking plug** using a heat gun.
- **Remove** the syphon **blanking plug** using the syphon 'C' spanner

Fig. 82: Measure the of depth of ice



(1) Depth marker
(2) Depth of ice in the turret

- Insert the service syphon into the syphon port.
- Slowly lower the syphon until a **resistance** is felt.
  » The extension tip will now be touching the ice.
- The **depth of ice in the syphon port** is shown in (pos 2).

2.3.4 Start the de-icing process:

- Open the magnet **bypass valve**.
- Turn on **helium regulator** and set the outlet pressure to **2 bar** on the gauge
- Wait **one minute**:
  » Unscrew the **syphon ring nut** enough to allow the syphon to be pushed gently down 10 to 15mm as the ice melts.
- Wait **one minute**:
  » Repeat the process - push the syphon down a further **10 to 15 mm**.

It is possible to melt **10 to 15 mm of ice per minute.** The process uses approximately one full (50 litre) cylinder of helium, taking thirty minutes to empty.

The **internal magnet pressure** will gradually rise during the de-ice process.

- Continue until the syphon has **engaged in the cone** either:
  » by feeling the **PTFE tip** hitting the **syphon cone**.
  » by reference to the mark made on the syphon **extension leg** . (➔ 3/Fig. 79 Page 80)
- Continue the de-icing for a further **ten minutes** then turn off the helium gas pressure at the cylinder regulator.

> The internal magnet pressure will rise quickly when there is a clear path through the ice into the helium vessel.

- When the **magnet pressure** has returned to atmosphere (**zero**), **CLOSE** the magnet **bypass valve**.
- **Remove** the **regulator line** from the syphon end adaptor.
- **Remove** the **NW25 flow restrictor** and adaptor from the end of the syphon.
- Fit the **syphon end cap**.
- Warm up the **syphon ring nut**.

2.3.5 Final Steps

If the magnet requires a **further LHe fill,** the syphon can be left in place.

- **Lift up** the syphon approximately **50 mm**:
  - » This will allow helium liquid to flow from the **standard syphon height.**
- **Re-tighten** the **syphon ring nut.**
- **Continue with helium filling** using standard **filling procedures.** (➔ Liquid helium transfer - top fill / Page 47)

3 Changes to Previous Version

Initial version.

There are no Hazard IDs in this document.

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